



Fiber products under toric flops and flips

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Abstract

Let Σ and Σ' be two refinements of a fan Σ_0 and $f: X_\Sigma \dashrightarrow X_{\Sigma'}$ be the birational map induced by $X_\Sigma \rightarrow X_{\Sigma_0} \leftarrow X_{\Sigma'}$. We show that the graph closure $\overline{\Gamma}_f$ is a *not necessarily normal* toric variety and we give a combinatorial criterion for its normality. In contrast to it, for f being a toric flop/flip, we show that the scheme-theoretic fiber product $X := X_\Sigma \times_{X_{\Sigma_0}} X_{\Sigma'}$ is in general *not toric*, though it is still *irreducible* and $X_{\text{red}} = \overline{\Gamma}_f$. A complete numerical criterion to ensure $X = X_{\text{red}}$ is given for 3-folds, which is fulfilled when X_Σ has at most terminal singularities. In this case, we further conclude that X is normal. The result turns out to be a key step to study the correspondence between Chow motives under toric flops and flips.

1 Introduction

For two schemes Y and Y' over a scheme S , the fiber product $Y \times_S Y'$ exists and is unique up to isomorphisms. It is natural to ask “Is the scheme-theoretic fiber product of two toric varieties still toric?” Unfortunately, the answer is “NO”. For example, let $\sigma = \text{Cone}(e_1, e_2)$ and $\sigma' = \text{Cone}(e_1, e_1 + e_2)$ which associate two affine toric varieties $U_\sigma = \text{Spec } \mathbb{C}[x^{e_1^\vee}, x^{e_2^\vee}]$ and $U_{\sigma'} = \text{Spec } \mathbb{C}[x^{e_1^\vee}, x^{e_2^\vee}, x^{e_1^\vee - e_2^\vee}]$. It is easy to see that

$$U_{\sigma'} \times_{U_\sigma} U_{\sigma'} = \text{Spec } \mathbb{C}[u, v, w, w'] / \langle u - vw, v(w - w') \rangle$$

and $\langle u - vw, v(w - w') \rangle$ is not a prime ideal, so $U_{\sigma'} \times_{U_\sigma} U_{\sigma'}$ is not a toric variety.

In birational geometry, the minimal model program plays an important role, in which the key ingredients consist of flops and flips, so it is a good choice to study toric fiber products

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under flops and flips. Actually the study of toric fiber products is related to the study of graph closures.

In this paper, we study the fundamental problem: *what is the difference between the graph closure and the fiber product under toric flops and flips*.¹ Notice that we use the definition of toric varieties by constructing them from a finite subset $\mathcal{A}$ of the lattice M of characters of the torus (see Sect. 2.1) and thus the toric varieties may be *not normal*. But if we regard $\mathcal{A}$ as a generating set in M and set $\sigma = \text{Cone}(\mathcal{A})^\vee \subseteq N_{\mathbb{R}}$ where N is the dual lattice of M , by [1], σ is a strongly convex rational polyhedral cone and then $\text{Spec}(\mathbb{C}[S_\sigma])$ is the normalization of our original toric variety constructed from $\mathcal{A}$.

Since a subscheme of a toric variety may not be a toric variety, the first thing we have to do is to study the toric structure of graph closures and fiber products under toric flops and flips. For the toric structure of graph closures, we can consider a more general birational map from two refinements of a fan. Indeed, given a fan Σ_0 in $N_{\mathbb{R}}$, a fan Σ refining Σ_0 will yield a toric morphism $X_\Sigma \rightarrow X_{\Sigma_0}$. In Theorem 3.1, let Σ and Σ' be two refinements of a fan Σ_0 and let $f: X_\Sigma \dashrightarrow X_{\Sigma'}$ be the birational map induced by $X_\Sigma \rightarrow X_{\Sigma_0} \leftarrow X_{\Sigma'}$. Then *the graph closure $\overline{\Gamma}_f$ is a toric variety*. Moreover, if we define the coarsest common refinement of Σ and Σ' over Σ_0 by

$$\tilde{\Sigma} := \{\sigma \cap \sigma' \mid \sigma \in \Sigma, \sigma' \in \Sigma' \text{ such that } \sigma, \sigma' \subseteq \sigma_0 \text{ for some } \sigma_0 \in \Sigma_0\}$$

in $N_{\mathbb{R}}$, then *the normalization of $\overline{\Gamma}_f$ is the toric variety $X_{\tilde{\Sigma}}$* .

The following is a combinatorial criterion for the normality of graph closures.

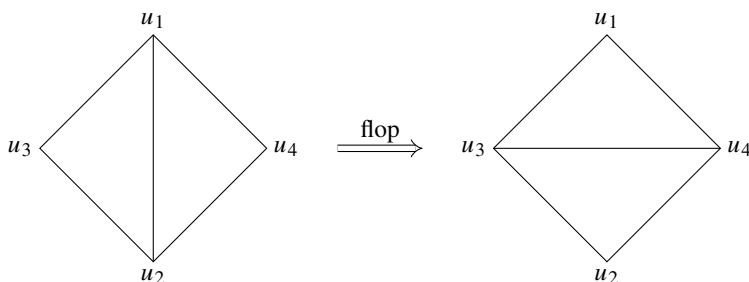
Theorem 1.1 (= Theorem 3.2) *The following statements are equivalent:*

- (i) $\overline{\Gamma}_f$ is normal.
- (ii) $\overline{\Gamma}_f$ is the toric variety associated with the fan $\tilde{\Sigma}$.
- (iii) For all (maximal cone) σ and σ' contained in some cone of Σ_0 ,

$$S_\sigma + S_{\sigma'} = S_{\sigma \cap \sigma'}.$$

Using the combinatorial criterion, in Corollary 3.3, we prove that if the toric variety X_Σ is smooth, then $\overline{\Gamma}_f$ is normal for a toric flop or flip f . Moreover, if $n = 3$ and $f: X_\Sigma \dashrightarrow X_{\Sigma'}$ is a toric flip or flop, then to achieve the normality of $\overline{\Gamma}_f$, we only need the smoothness of one affine piece of X_Σ .

The geometric picture of a toric flip is outlined as follows. For smooth toric 3-folds, Danilov in late 70's proved that one can move the fan Σ to Σ' by a sequence of elementary flops. In each step, it corresponds to the move



where the primitive vectors u_i lie in a plane and $u_1 + u_2 = u_3 + u_4$. Geometrically this is the blowing-up of a $(-1, -1)$ rational curve in a 3-fold then followed by a blowing-down of the

¹ For our terminology of toric flops and flips, see Remark 2.10.

exceptional divisor $\mathbb{P}^1 \times \mathbb{P}^1$ in another direction, that is, it is an easy ordinary flop. Reid in early 80's generalized the above elementary move to higher dimensions: let $u_1, \dots, u_{n+1}$ be primitive vectors in $N = \mathbb{Z}^n$ such that σ_n and σ_{n+1} be two top dimensional cones intersect along the face cone $\tau = \text{Cone}(u_1, \dots, u_{n-1})$, where $\sigma_j := \text{Cone}(u_1, \dots, \hat{u}_j, \dots, u_{n+1})$. Let the linear relation between u_i 's be

$$b_1 u_1 + \dots + b_n u_n + u_{n+1} = 0,$$

which is called a wall relation. Here we set $b_{n+1} = 1$ and we must have $b_n > 0$ since u_n and u_{n+1} lie in opposite sides of τ . Reordering $u_1, \dots, u_{n-1}$ we may assume that $b_i < 0$ for $1 \leq i \leq \alpha$, $b_i = 0$ for $\alpha + 1 \leq i \leq \beta$ and $b_i > 0$ for $\beta + 1 \leq i \leq n + 1$. Notice that $0 \leq \alpha \leq \beta \leq n - 1$.

The case one can perform “elementary move” is when $\alpha \geq 2$. Then there are two different decompositions of $\sigma_0 := \text{Cone}(u_1, \dots, u_{n+1})$:

$$\sigma_0 = \bigcup_{\beta+1 \leq j \leq n+1} \sigma_j = \bigcup_{1 \leq j \leq \alpha} \sigma_j.$$

Our original two cones are in the first decomposition and the second decomposition will give us the local construction of a *toric flip*, whose global definition is given in Sect. 2.3. When $u_1, \dots, u_{n+1}$ lie in an affine hyperplane of $N_{\mathbb{Q}}$, it leads to a *toric flop*.

Next, for toric flops and flips, we study the toric structure of their fiber products. In Proposition 4.1, we find that the fiber product $X := X_{\Sigma} \times_{X_{\Sigma_0}} X_{\Sigma'}$ is a normal toric variety if and only if the scheme X is (1) irreducible, (2) reduced and (3) the graph closure $\bar{\Gamma}_f$ is a normal toric variety. The condition (3) is studied in Sect. 3 mentioned as above. For the condition (1), we show that X is always irreducible under a toric flop or flip. Moreover, the reduced scheme X_{red} with respect to X is the toric variety $\bar{\Gamma}_f$. In particular, if X is a normal toric variety, then $X = \bar{\Gamma}_f = X_{\Sigma}$.

Theorem 1.2 (= Theorem 4.3) *Let $f: X_{\Sigma} \dashrightarrow X_{\Sigma'}$ be a toric flip via toric morphisms $X_{\Sigma} \rightarrow X_{\Sigma_0} \leftarrow X_{\Sigma'}$ (2.10) and $X = X_{\Sigma} \times_{X_{\Sigma_0}} X_{\Sigma'}$ be the fiber product. Then we have:*

- (i) *The reduced scheme X_{red} associated to X is a (not necessarily normal) toric variety.*
- (ii) *The normalization of X_{red} is the toric variety X_{Σ} .*

The condition (2) of Proposition 4.1 is the biggest cause of uncertainty. So far, some useful criteria are given only for 3-dimensional case. Since the property of being reduced is local, we usually assume X_{Σ_0} is an affine toric variety U_{σ_0} defined by σ_0 and $U_{ji} := U_{\sigma_j} \times_{U_{\sigma_0}} U_{\sigma_i}$. Also, we can assume $b_i \in \mathbb{Z}$, $\gcd(b_1, b_2, b_3, b_4) = 1$, and $b_4 = 1$ by the assumption U_{σ_4} is smooth. We provide a numerical criterion for the reduced property on the affine piece U_{31} .

Theorem 1.3 (= Lemma 4.5) *Let $\{a\}_b$ denote the remainder of a divided by b . If $g = \gcd(b_1, b_2) > 0$ and $b_i = -gb'_i$ for $i = 1, 2$, then the following statements are equivalent:*

- (i) U_{31} is reduced.
- (ii) For all $0 \leq \lambda \leq b'_1 b'_2$, there exists a non-negative integer $y \leq \lambda/b'_1$ such that

$$\{g\lambda\}_{b_3} \geq g \cdot \{\lambda - b'_1 y\}_{b'_2}. \quad (1.1)$$

Finally, we show that the reduced property of the special affine piece implies the reduced property of the whole X .

Theorem 1.4 (= Theorem 4.7) *X is reduced if and only if U_{31} is reduced.*

As an application, we get the following expected result for 3-dimensional case.

Theorem 1.5 (= Theorem 5.1) *If X_Σ is a 3-dimensional simplicial toric variety with at worst terminal singularities, then the fiber product $X = X_\Sigma \times_{X_{\Sigma_0}} X_{\Sigma'}$ is the toric variety $X_{\tilde{\Sigma}}$.*

A generalized version of Lemma 4.5 can also help us to get a criterion for the smooth higher dimensional case.

Theorem 1.6 (= Theorem 5.5) *Assume that X_Σ is smooth of dimension n and X_{Σ_0} is affine with the wall relation (5.1). Then the fiber product $X = X_\Sigma \times_{X_{\Sigma_0}} X_{\Sigma'}$ is the toric variety $X_{\tilde{\Sigma}}$ if and only if*

$$b_i \mid b_j \text{ or } b_j \mid b_i$$

for any $i, j \in J_-$.

This work is also motivated by the conjecture given by Wang in 2001 (ref. [10]), which said that for K -equivalent manifolds under birational map $f: X \dashrightarrow X'$, there is a *naturally attached* correspondence $T \in A^{\dim X}(X \times X')$ of the form $T = \tilde{\Gamma}_f + \sum_i T_i$ with $\tilde{\Gamma}_f \subseteq X \times X'$ the cycle of graph closure of f and with T_i 's being certain degenerate correspondences (i.e. T_i has positive dimensional fibers when projecting to X or X') such that T is an isomorphism of Chow motives. In [6], the authors showed that for an ordinary $\mathbb{P}^r$ flop $f: X \dashrightarrow X'$, the graph closure $[\tilde{\Gamma}_f] \in A^*(X \times X')$ identifies the Chow motives $\hat{X}$ of X and $\hat{X}'$ of X' . More generally, for f an ordinary (r, r') flip with $r \leq r'$, the graph closure $[\tilde{\Gamma}_f] \in A^*(X \times X')$ identifies the Chow motive $\hat{X}$ of X as a sub-motive of $\hat{X}'$ which preserves also the Poincaré pairing on cohomology groups.

In toric case, we have the following observation which is well-known for experts. (For example, cf. [1] or [5].)

Remark 1.7 (= Proposition A.1 + Theorem A.2)

- (i) Any two K -equivalent simplicial terminal toric varieties can be connected to each other by a sequence of toric flops.
- (ii) Any smooth toric flop is an ordinary flop.

We would like to give a simple proof in Appendix A by supplementing Reid's theory on toric minimal model program. Now, together with the result in [6], if K -equivalent toric manifolds are connected by smooth flops, then they admit canonically isomorphic integral cohomology groups via the graph closure. To study the conjecture for toric manifolds in [10], it is important to know for a general toric flop whether the graph closure still provides a canonical isomorphism between the integral cohomology groups, or we need extra degenerate correspondence in their fiber product.

As an application, under a 3-dimensional terminal toric flop f , its fiber product is equal to the graph closure and thus is expected to give the equivalence of their Chow motives.

2 Preliminaries

We begin by recalling the basic notions of toric varieties and fixing our notation. For further details, see [1] and [3].

A toric variety is a (not necessarily normal) variety X containing an algebraic torus as a Zariski open subset, together with an algebraic action of the torus on X that extends the natural action of the torus on itself. Let $M \simeq \mathbb{Z}^n$ be the lattice of characters of the torus,

with dual lattice $N := \text{Hom}_{\mathbb{Z}}(M, \mathbb{Z})$. Then the torus is canonically isomorphic to $N \otimes_{\mathbb{Z}} \mathbb{C}^*$, denoted by T_N . Throughout this paper, we fix the lattices M and N of rank n , and their $\mathbb{R}$ -linear extensions $M_{\mathbb{R}} = M \otimes \mathbb{R}$ and $N_{\mathbb{R}} = N \otimes \mathbb{R}$.

2.1 Affine toric varieties

Given a set $\mathcal{A} = \{m_1, \dots, m_s\} \subseteq M$, we get characters $\chi^{m_i} : T_N \rightarrow \mathbb{C}^*$, the affine semigroup $S := \mathbb{Z}_{\geq 0} \mathcal{A} \subseteq M$ and the *semigroup algebra* $\mathbb{C}[S] := \mathbb{C}[\chi^{m_1}, \dots, \chi^{m_s}]$ with multiplication induced by the semigroup structure of S . This gives rise to an affine toric variety $\text{Spec}(\mathbb{C}[S])$, which is the Zariski closure of the image of $T_N \rightarrow \mathbb{C}^s$ defined by characters χ^{m_i} . In particular, the dimension of the affine toric variety is the rank of $\mathbb{Z}\mathcal{A}$.

Example 2.1 The affine semigroup $\mathbb{Z}_{\geq 0}^s \subseteq \mathbb{Z}^s$ gives the polynomial ring

$$\mathbb{C}[\mathbb{Z}_{\geq 0}^s] = \mathbb{C}[x_1, \dots, x_s],$$

where $x_i = \chi^{e_i}$ and $\{e_1, \dots, e_s\}$ is the standard basis of $\mathbb{Z}^s$.

In what follows, we define $\mathbb{Z}^{\mathcal{A}} = \bigoplus_{i=1}^s \mathbb{Z}e_{m_i} \cong \mathbb{Z}^s$ and also write $\mathbb{C}[\mathbb{Z}_{\geq 0}^{\mathcal{A}}] = \mathbb{C}[\mathbb{Z}_{\geq 0}^s]$ with $\chi^{e_{m_i}} = x_i$ for $\mathcal{A} = \{m_1, \dots, m_s\}$ when there is no danger of confusion. An inclusion $\mathcal{A} \subseteq \mathcal{B}$ induces a natural homomorphism $\mathbb{C}[\mathbb{Z}_{\geq 0}^{\mathcal{A}}] \rightarrow \mathbb{C}[\mathbb{Z}_{\geq 0}^{\mathcal{B}}]$ of polynomial rings.

Definition 2.2 The *toric ideal* $I_{\mathcal{A}}$ of the affine toric variety $\text{Spec}(\mathbb{C}[S])$ is defined by the kernel of the surjective $\mathbb{C}$ -algebra homomorphism $\mathbb{C}[\mathbb{Z}_{\geq 0}^{\mathcal{A}}] \rightarrow \mathbb{C}[S]$ where $x_i \mapsto \chi^{m_i}$.

We will use the convention in this paper that m_{α} denotes the lattice point $\sum_{i=1}^r a_i m_i$ for any given set $\mathcal{S} = \{m_1, \dots, m_r\} \subseteq M$ and $\alpha = (a_1, \dots, a_r) \in \mathbb{Z}^r = \mathbb{Z}^{\mathcal{S}}$. It induces a map of character lattices

$$\mathbb{Z}^{\mathcal{A}} = \mathbb{Z}^s \rightarrow M$$

that sends α to the lattice point m_{α} . Let $L_{\mathcal{A}}$ be defined by the following exact sequence

$$0 \rightarrow L_{\mathcal{A}} \rightarrow \mathbb{Z}^{\mathcal{A}} \rightarrow M.$$

Then the toric ideal of $\text{Spec}(\mathbb{C}[S])$ is the prime ideal

$$I_{\mathcal{A}} = \langle x^{\alpha} - x^{\beta} \mid \alpha, \beta \in \mathbb{Z}_{\geq 0}^s \text{ and } \alpha - \beta \in L_{\mathcal{A}} \rangle \quad (2.1)$$

where $x^{\gamma} = x_1^{\gamma_1} x_2^{\gamma_2} \cdots x_s^{\gamma_s}$ for any $\gamma \in \mathbb{Z}_{\geq 0}^s$.

Example 2.3 Given a rational polyhedral cone $\sigma \subseteq N_{\mathbb{R}}$, the lattice points

$$S_{\sigma} := \sigma^{\vee} \cap M \subseteq M$$

form a semigroup. It is finitely generated by Gordan's Lemma. Therefore this affine semigroup gives us an affine toric variety

$$U_{\sigma} := \text{Spec}(\mathbb{C}[S_{\sigma}]).$$

Furthermore, the cone σ is strongly convex if and only if $\dim U_{\sigma} = \dim \sigma^{\vee} = \dim_{\mathbb{R}} M_{\mathbb{R}}$.

In fact, $\text{Spec}(\mathbb{C}[S])$ is not necessarily normal. If we regard $\mathcal{A}$ as a generating set in M and set $\sigma = \text{Cone}(\mathcal{A})^{\vee} \subseteq N_{\mathbb{R}}$. By [1, Proposition 1.3.8], σ is a strongly convex rational polyhedral cone and the inclusion $\mathbb{C}[S] \subseteq \mathbb{C}[S_{\sigma}]$ induces a morphism $U_{\sigma} \rightarrow \text{Spec}(\mathbb{C}[S])$ that is the normalization map of $\text{Spec}(\mathbb{C}[S])$. It gives us an important fact.

Fact 2.4 $\text{Spec}(\mathbb{C}[S_\sigma])$ is the normalization of $\text{Spec}(\mathbb{C}[S])$. Actually, the semigroup S_σ is the saturation of S .

The following lemma is used in the proof of Theorem 3.1.

Lemma 2.5 Let σ be a strongly convex rational polyhedral cone in $N_{\mathbb{R}}$. For any $m_1, m_2 \in M$, we can find a lattice point $m_0 \in S_\sigma$ such that $m_0 + m_i \in S_\sigma$ for $i = 1, 2$.

Proof Notice that $\dim \sigma^\vee = \dim_{\mathbb{R}} M_{\mathbb{R}}$ since σ is strongly convex. Fix a lattice point m' in the interior $\text{int}(\sigma^\vee)$ of $\sigma^\vee$. Since $m' + (1/\ell)m_i \in \text{int}(\sigma^\vee)$ for a sufficiently large integer ℓ , $\ell m' + m_i \in S_\sigma$ for $i = 1, 2$. $\square$

Assume that σ, σ' and σ_0 are three rational polyhedral cones in $N_{\mathbb{R}}$ such that σ_0 contains σ and σ' . Let $\mathcal{A}_0 \subseteq M$ be a generating set of S_{σ_0} and similarly for $\mathcal{A}$ and $\mathcal{A}'$. Since $\sigma^\vee \cap (\sigma')^\vee$ contains $\sigma_0^\vee$, we may assume that $\mathcal{A} \cap \mathcal{A}' \supseteq \mathcal{A}_0$. Consider the affine semigroup

$$S_{\text{split}} := \mathbb{Z}_{\geq 0}^{\mathcal{A}_0} \oplus \mathbb{Z}_{\geq 0}^{\mathcal{A} \setminus \mathcal{A}_0} \oplus \mathbb{Z}_{\geq 0}^{\mathcal{A}' \setminus \mathcal{A}_0}. \quad (2.2)$$

Via $\mathbb{C}[\mathbb{Z}_{\geq 0}^{\mathcal{A}}]$ and $\mathbb{C}[\mathbb{Z}_{\geq 0}^{\mathcal{A}'}]$ as subrings of $\mathbb{C}[S_{\text{split}}]$, by abuse of notation, we still write $I_{\mathcal{A}}$ and $I_{\mathcal{A}'}$ for the corresponding ideals in $\mathbb{C}[S_{\text{split}}]$. The following lemma characterizes binomials of $\mathbb{C}[S_{\text{split}}]$ which belong to the sum $I_{\mathcal{A}} + I_{\mathcal{A}'}$ of toric ideals, which is used in the proof of Lemma 4.5.

Lemma 2.6 For $\alpha_0, \beta_0 \in \mathbb{Z}_{\geq 0}^{\mathcal{A}}$, $\alpha, \beta \in \mathbb{Z}_{\geq 0}^{\mathcal{A} \setminus \mathcal{A}_0}$ and $\alpha', \beta' \in \mathbb{Z}_{\geq 0}^{\mathcal{A}' \setminus \mathcal{A}_0}$, the binomial

$$x^{\alpha_0 + \alpha + \alpha'} - x^{\beta_0 + \beta + \beta'} \in I_{\mathcal{A}} + I_{\mathcal{A}'} \subseteq \mathbb{C}[S_{\text{split}}] \quad (2.3)$$

if and only if there exists a sequence $\{(\gamma_{0i}, \gamma_i, \gamma'_i)\}_{i=0}^m \subseteq S_{\text{split}}$ such that

$$(\gamma_{01}, \gamma_1, \gamma'_1) = (\alpha_0, \alpha, \alpha'), \quad (\gamma_{0m}, \gamma_m, \gamma'_m) = (\beta_0, \beta, \beta'),$$

and for each $1 \leq i \leq m-1$, either

$$\gamma_i = \gamma_{i+1} \text{ and } (\gamma_{0i} + \gamma'_i) - (\gamma_{0(i+1)} + \gamma'_{i+1}) \in L_{\mathcal{A}'}, \text{ or} \quad (2.4)$$

$$\gamma'_i = \gamma'_{i+1} \text{ and } (\gamma_{0i} + \gamma_i) - (\gamma_{0(i+1)} + \gamma_{i+1}) \in L_{\mathcal{A}} \quad (2.5)$$

holds.

Proof The necessary part is clear, since

$$x^{\gamma_{0i} + \gamma_i + \gamma'_i} - x^{\gamma_{0(i+1)} + \gamma_{i+1} + \gamma'_{i+1}} = \begin{cases} x^{\gamma_i} (x^{\gamma_{0i} + \gamma'_i} - x^{\gamma_{0(i+1)} + \gamma'_{i+1}}) \in I_{\mathcal{A}'} & \text{if (2.4) holds} \\ x^{\gamma'_i} (x^{\gamma_{0i} + \gamma_i} - x^{\gamma_{0(i+1)} + \gamma_{i+1}}) \in I_{\mathcal{A}} & \text{if (2.5) holds} \end{cases}$$

for each $1 \leq i \leq m-1$.

Conversely, suppose that (2.3) holds. We write

$$x^{\alpha_0 + \alpha + \alpha'} - x^{\beta_0 + \beta + \beta'} = \sum_{k=1}^{\ell} c_k (x^{\gamma_k^+} - x^{\gamma_k^-}),$$

where $c_k \in \mathbb{C}$ and $\{(\gamma_k^+, \gamma_k^-)\}_{k=1}^{\ell}$ are distinct pairs satisfying (2.4) or (2.5) under the decomposition (2.2). Consider a weighed directed graph G with vertices $\{\gamma_k^+, \gamma_k^- \mid 1 \leq k \leq \ell\}$, directed edges from γ_k^- to γ_k^+ with weight $\text{wt}(\gamma_k^-, \gamma_k^+) = c_k$. Define the total degree at vertex v by

$$\sum_{(u,v) \in E(G)} \text{wt}(u, v) - \sum_{(v,u) \in E(G)} \text{wt}(v, u).$$

Then the total degree at $\alpha_0 + \alpha + \alpha'$ is 1, at $\beta_0 + \beta + \beta'$ is -1 , and at other vertices are 0. Since the sum of total degrees over the vertices in a connected component of G is zero, we conclude that $\alpha_0 + \alpha + \alpha'$ and $\beta_0 + \beta + \beta'$ are in the same connected component of G . Consequently, they can be connected by undirected edges, and the vertices along this path form the desired sequence, since the two vertices connected by the edge in G satisfy (2.4) or (2.5). $\square$

2.2 Fans

For any fan Σ in $N_{\mathbb{R}}$, there is a corresponding normal toric variety X_{Σ} on which the torus T_N acts naturally. If a toric variety is normal, then it always comes from a fan Σ in $N_{\mathbb{R}}$ by Sumihiro's Theorem [9]. The assignment $\Sigma \mapsto X_{\Sigma}$ yields an equivalence of categories between the category of fans with morphisms of fans and the category of normal toric varieties with toric morphisms.

Given a fan Σ_0 in $N_{\mathbb{R}}$, a fan Σ *refines* Σ_0 if every cone of Σ is contained in a cone of Σ_0 and $|\Sigma| = |\Sigma_0|$. This yields a toric morphism $X_{\Sigma} \rightarrow X_{\Sigma_0}$ whose restriction map on T_N is the identity map.

Definition 2.7 Let Σ and Σ' be two refinements of Σ_0 . We define the coarsest common refinement of Σ and Σ' over Σ_0 by

$$\tilde{\Sigma} := \{\sigma \cap \sigma' \mid \sigma \in \Sigma, \sigma' \in \Sigma' \text{ such that } \sigma, \sigma' \subseteq \sigma_0 \text{ for some } \sigma_0 \in \Sigma_0\} \quad (2.6)$$

in $N_{\mathbb{R}}$.

Lemma 2.8 *The coarsest common refinement $\tilde{\Sigma}$ of Σ and Σ' over Σ_0 is a fan.*

Proof To show that $\tilde{\Sigma}$ is a fan, first consider cones $\sigma \in \Sigma$ and $\sigma' \in \Sigma'$. Note that $\sigma \cap \sigma'$ is a rational polyhedral cone, since σ and σ' are the intersections of finitely many half-spaces in $N_{\mathbb{R}}$. Additionally, $\sigma \cap \sigma'$ is strongly convex, since σ is strongly convex.

Next consider a face $\tilde{\tau}$ of $\sigma \cap \sigma'$, say $\tilde{\tau} = (\sigma \cap \sigma') \cap \text{Cone}(m)^{\perp}$ for some $m \in (\sigma \cap \sigma')^{\vee}$. We can write $\tilde{m} = m + m'$ under the decomposition

$$(\sigma \cap \sigma')^{\vee} = (\sigma^{\vee\vee} \cap \sigma'^{\vee\vee})^{\vee} = (\sigma^{\vee} + \sigma'^{\vee})^{\vee\vee} = \sigma^{\vee} + \sigma'^{\vee}. \quad (2.7)$$

Then $\tau = \sigma \cap \text{Cone}(z)^{\perp}$ and $\tau' = \sigma' \cap \text{Cone}(z')^{\perp}$ are faces of σ and σ' respectively. Since $\langle m, u \rangle, \langle m', u \rangle \geq 0$ for all $u \in \sigma \cap \sigma'$, the equation $\langle \tilde{m}, u \rangle = \langle m, u \rangle + \langle m', u \rangle$ implies that

$$\tilde{\tau} = \tau \cap \tau' \in \tilde{\Sigma}.$$

The same argument also implies that any face of $\sigma \cap \sigma'$ has the form $\tau \cap \tau'$, where τ and τ' are the faces of σ and σ' respectively.

Finally, we need to show that the intersection of any two cones $\sigma_1 \cap \sigma'_1$ and $\sigma_2 \cap \sigma'_2$ of $\tilde{\Sigma}$ is a face of each, where $\sigma_i \in \Sigma$ and $\sigma'_i \in \Sigma'$. Since $\sigma_1 \cap \sigma_2$ and $\sigma'_1 \cap \sigma'_2$ are faces of σ_i and σ'_i respectively, we deduce that

$$(\sigma_1 \cap \sigma'_1) \cap (\sigma_2 \cap \sigma'_2) = (\sigma_1 \cap \sigma_2) \cap (\sigma'_1 \cap \sigma'_2)$$

is the face of $\sigma_i \cap \sigma'_i$ for $i = 1, 2$. So we conclude that $\tilde{\Sigma}$ forms a fan in $N_{\mathbb{R}}$ $\square$

We note that $\tilde{\Sigma}$ is the fiber product of Σ and Σ' over Σ_0 in the category of fans, so it is an important ingredient to study the fiber product of toric morphisms.

2.3 Wall relations and toric flips

In this section, we recall some basic results about toric flips. For details, please refer to [1, Chapter 15], [7, Chapter 14] or the paper [8].

Let X_Σ be a simplicial semiprojective toric variety and let $\mathcal{R} \subseteq \overline{\text{NE}}(X_\Sigma)$ be an extremal ray of its Mori cone. Reid showed that there is an extremal contraction

$$\phi_{\mathcal{R}}: X_\Sigma \rightarrow X_{\Sigma_0}$$

such that Σ refines Σ_0 and X_{Σ_0} is semiprojective.

The idea of the construction for $\phi_{\mathcal{R}}$ is briefly stated as follows. The extremal ray $\mathcal{R}$ is generated by the curve class of an orbit closure $V(\tau)$ of an $(n-1)$ -dimensional cone $\tau \in \Sigma$ and τ is called a *wall*. Roughly speaking, Σ_0 is obtained from the fan Σ by “removing” all walls $\tau \in \Sigma$ with $[V(\tau)] \in \mathcal{R}$.

To see the local picture, we pick a wall $\tau = \text{Cone}(u_1, \dots, u_{n-1})$ with $[V(\tau)] \in \mathcal{R}$. Since Σ is simplicial, the wall τ separates two n -dimensional cones

$$\begin{aligned}\sigma_{n+1} &= \text{Cone}(u_1, \dots, u_n), \\ \sigma_n &= \text{Cone}(u_1, \dots, u_{n-1}, u_{n+1})\end{aligned}$$

in Σ . The primitive vectors $u_1, \dots, u_{n+1}$ of $\rho_1, \dots, \rho_{n+1}$ in $\Sigma(1)$ span $N_{\mathbb{R}} \simeq \mathbb{R}^n$ and there is a nontrivial linear relation

$$\sum_{i=1}^{n+1} b_i u_i = 0 \quad (2.8)$$

over $\mathbb{Q}$, which is unique up to multiplication by a nonzero rational number and is called a *wall relation*. Here, we may assume $b_{n+1} > 0$. If $D_i = V(\rho_i)$ for $i = 1, \dots, n+1$, then we have that

$$D_i \cdot V(\tau) = \frac{b_i}{b_{n+1}} D_{n+1} \cdot V(\tau)$$

from the wall relation. For $\rho \in \Sigma(1)$ with its primitive vector $u_\rho \notin \{u_1, \dots, u_{n+1}\}$, $D_\rho \cdot \mathcal{R} = 0$ since ρ and τ can not form a cone in Σ . We conclude that the sets

$$\begin{aligned}\{u_i \mid b_i < 0\} &= \{u_\rho \mid \rho \in \Sigma(1), D_\rho \cdot \mathcal{R} < 0\}, \\ \{u_i \mid b_i > 0\} &= \{u_\rho \mid \rho \in \Sigma(1), D_\rho \cdot \mathcal{R} > 0\}\end{aligned} \quad (2.9)$$

are independent of the choice of the wall τ with $[V(\tau)] \in \mathcal{R}$.

Furthermore, since X_Σ is simplicial, we have the following exact sequence

$$0 \rightarrow M \rightarrow \mathbb{Z}^{\Sigma(1)} \rightarrow \text{Cl}(X_\Sigma) \rightarrow 0,$$

defined by $m \mapsto (\langle m, u_\rho \rangle)_\rho$ and $e_\rho \mapsto D_\rho$. Its dual sequence is

$$0 \rightarrow N_1(X_\Sigma)_{\mathbb{R}} \rightarrow \mathbb{R}^{\Sigma(1)} \rightarrow N_{\mathbb{R}} \rightarrow 0,$$

so we can identify $N_1(X_\Sigma)_{\mathbb{R}}$ with linear relations of primitive vectors in $\Sigma(1)$ and thus the wall relation (2.8) is uniquely determined by the extremal ray $\mathcal{R}$ up to multiplication by a nonzero rational number.

In [8], Reid had shown that Σ_0 is a nonsimplicial fan if and only if $\phi_{\mathcal{R}}$ is a small birational contraction. In this case, $|\{\rho \in \Sigma(1) \mid D_\rho \cdot \mathcal{R} < 0\}| > 1$ and

$$\sigma_0 := \text{Cone}(u_1, \dots, u_n, u_{n+1})$$

is a non-simplicial n -dimensional cone in Σ_0 . Also, there is a commutative diagram of birational toric morphisms

$$\begin{array}{ccc} X_\Sigma & & X_{\Sigma'} \\ & \searrow \phi_{\mathcal{R}} & \swarrow \phi' \\ & X_{\Sigma_0} & \end{array} \quad (2.10)$$

where Σ' is given by another subdivision of Σ_0 . The birational map

$$f := (\phi')^{-1} \circ \phi_{\mathcal{R}} : X_\Sigma \dashrightarrow X_{\Sigma'} \quad (2.11)$$

is called the *toric flip* of $\phi_{\mathcal{R}}$, and is called the *toric flop* of $\phi_{\mathcal{R}}$ if $K_{X_\Sigma} \cdot \mathcal{R} = 0$.

To say more about the fan Σ' , for the wall τ as above, we get the affine toric subvariety $U_{\sigma_0} \subseteq X_{\Sigma_0}$. From the wall relation (2.8), we define the sets

$$J_- = \{i \mid b_i < 0\}, J_0 = \{i \mid b_i = 0\}, J_+ = \{i \mid b_i > 0\}$$

and the cones

$$\sigma_J = \text{Cone}(u_i \mid i \in J) \text{ for } J \subseteq \{1, \dots, n+1\}.$$

Via two subdivisions of σ_0 , Σ contains the cones $\{\sigma_J \mid J_+ \not\subseteq J\}$ and Σ' contains the cones $\{\sigma_J \mid J_- \not\subseteq J\}$. Also, the exceptional loci of $\phi_{\mathcal{R}}$ and ϕ' over U_{σ_0} are $V(\sigma_{J_-})$ and $V(\sigma_{J_+})$ which map onto $V(\sigma_{J_- \cup J_+})$, $\text{codim } V(\sigma_{J_\pm}) = |J_\pm| \geq 2$ and $\dim V(\sigma_{J_- \cup J_+}) = |J_0|$.

As mentioned in (2.9), $\{u_i \mid i \in J_- \cup J_+\}$ is the set of primitive vectors u_ρ of ρ in $\Sigma(1)$ such that $D_\rho \cdot \mathcal{R} \neq 0$. Therefore, if $\sigma_{\text{exc}} := \text{Cone}(u_\rho \mid \mathcal{R} \cdot D_\rho \neq 0) \in \Sigma_0$, then every $\sigma_0 \in \Sigma_0(n) \setminus \Sigma(n)$ comes from

$$\text{Star}(\sigma_{\text{exc}}) := \{\sigma_0 \in \Sigma_0(n) \mid \sigma_{\text{exc}} < \sigma_0\},$$

which is obtained by “removing” walls. Hence,

$$\text{Exc}(\phi_{\mathcal{R}}) = V(\text{Cone}(u_\rho \mid D_\rho \cdot \mathcal{R} < 0)) \quad \text{and} \quad \phi_{\mathcal{R}}(\text{Exc}(\phi_{\mathcal{R}})) = V(\sigma_{\text{exc}}).$$

In the subsequent sections, many problems can be checked locally, so we may fix $n+1$ vectors $u_1, \dots, u_{n+1}$ from a wall τ together with the wall relation (2.8).

The terminology used for toric flips and toric flops coincides with the usual one in the minimal model program.

Definition 2.9 Let (X, Δ) be a log canonical pair. A projective morphism $\phi : X \rightarrow Z$ between normal varieties is a $(K_X + \Delta)$ -*flipping contraction* if

- (1) X is $\mathbb{Q}$ -factorial and Δ is an $\mathbb{R}$ -divisor,
- (2) ϕ is a small birational morphism of relative Picard number 1,
- (3) $-(K_X + \Delta)$ is ϕ -ample.

A $(K_X + \Delta)$ -flipping contraction is a *flopping contraction* if K_X is numerically relatively trivial.

Remark 2.10 Recall that, following [8], a toric flip in this paper means a wall relation (2.8) with $|J_-| > 1$. Equivalently, it corresponds to the small birational contraction $\phi_{\mathcal{R}} : X_\Sigma \rightarrow X_{\Sigma_0}$ associated with an extremal ray $\mathcal{R}$ and the diagram (2.10).

Moreover, by [7, Theorem 14-3-3] (see also [1, p. 781]), there exists a torus invariant boundary $\mathbb{Q}$ -divisor Δ such that (X_Σ, Δ) is klt and $(K_{X_\Sigma} + \Delta) \cdot \mathcal{R} < 0$. Hence $\phi_{\mathcal{R}}$ is a $(K_{X_\Sigma} + \Delta)$ -flipping contraction in the sense of Definition 2.9, and it is a flopping contraction if and only if it gives a toric flop in our sense. Notably, this formulation does not require the condition $K_{X_\Sigma} \cdot \mathcal{R} < 0$ typically found in the classical Minimal Model Program. Indeed, the sign of $K_{X_\Sigma} \cdot \mathcal{R}$ has the same sign as $-\sum_{i=1}^{n+1} b_i$.

3 Graph closures

In this section, we will investigate the graph closure $\overline{\Gamma}_f$ of a toric birational map f which is endowed with its reduced subscheme structure.

Theorem 3.1 *Let Σ and Σ' be two refinements of a fan Σ_0 and let $f: X_\Sigma \dashrightarrow X_{\Sigma'}$ be the birational map induced by $X_\Sigma \rightarrow X_{\Sigma_0} \leftarrow X_{\Sigma'}$. Then we have:*

- (1) *The graph closure $\overline{\Gamma}_f$ is a (not necessarily normal) toric variety.*
- (2) *The normalization of $\overline{\Gamma}_f$ is the toric variety $X_{\tilde{\Sigma}}$ defined by the fan (2.6).*

In particular, the normalization of $\overline{\Gamma}_f$ is the fiber product in the category of normal toric varieties for such a birational map f .

Proof The questions are local, so we fix a cone $\sigma \cap \sigma' \in \tilde{\Sigma}$ and the open set $U_{\sigma \cap \sigma'}$. By (2.6), there is a cone $\sigma_0 \in \Sigma_0$ such that σ and σ' are contained in σ_0 . Let $\mathcal{A}_0 \subseteq M$ be a generating set of S_{σ_0} , and similarly for $\mathcal{A}$ and $\mathcal{A}'$. Since $\sigma^\vee \cap (\sigma')^\vee$ contains $\sigma_0^\vee$, we may assume that $\mathcal{A} \cap \mathcal{A}' \supseteq \mathcal{A}_0$.

First, we deal with the simple case that $\mathcal{A} \cap \mathcal{A}' = \mathcal{A}_0$. Set $\mathcal{B} = \mathcal{A} \cup \mathcal{A}'$. According to $\mathcal{A} \cap \mathcal{A}' \supseteq \mathcal{A}_0$, it follows that the fiber product $U_\sigma \times_{U_{\sigma_0}} U_{\sigma'}$ is the spectrum of the ring

$$A := \mathbb{C}[\mathbb{Z}_{\geq 0}^{\mathcal{A}}]/I_{\mathcal{A}} \otimes_{\mathbb{C}[\mathbb{Z}_{\geq 0}^{\mathcal{A}_0}]/I_{\mathcal{A}_0}} \mathbb{C}[\mathbb{Z}_{\geq 0}^{\mathcal{A}'}]/I_{\mathcal{A}'} \simeq \mathbb{C}[\mathbb{Z}_{\geq 0}^{\mathcal{B}}]/I_{\mathcal{A}} + I_{\mathcal{A}'}, \quad (3.1)$$

and $T_N \times_{T_N} T_N = \text{Spec } A_a$ where $a := \prod_{m \in \mathcal{A}_0} x_m$. Denote by $\iota: A \rightarrow A_a$ the canonical ring homomorphism. Since $T_N \times_{T_N} T_N \simeq T_N$ is integral, the zero ideal 0_{A_a} is a radical ideal. Then the graph closure $\overline{\Gamma}_f = \overline{T_N \times_{T_N} T_N}$ in $\text{Spec } A$ is defined by the ideal

$$\bigcap_{a \notin \mathfrak{p} \in \text{Spec } A} \mathfrak{p} = \iota^{-1}(\sqrt{0_{A_a}}) = \iota^{-1}(0_{A_a}).$$

Note that the toric (prime) ideal $I_{\mathcal{B}}$ contains $I_{\mathcal{A}} + I_{\mathcal{A}'}$. Then (1) follows from the claim that $\overline{\Gamma}_f \cap \text{Spec } A = \mathbf{V}(I_{\mathcal{B}})$ in $\text{Spec } A$. Indeed, using (2.1) and the identification

$$\mathbb{Z}^{\mathcal{B}} = \mathbb{Z}^{\mathcal{A}_0} \oplus \mathbb{Z}^{\mathcal{A} \setminus \mathcal{A}_0} \oplus \mathbb{Z}^{\mathcal{A}' \setminus \mathcal{A}_0}, \quad (3.2)$$

we pick

$$(\alpha_0 + \alpha + \alpha') - (\beta_0 + \beta + \beta') \in L_{\mathcal{B}} \quad (3.3)$$

where $\alpha_0, \beta_0 \in \mathbb{Z}_{\geq 0}^{\mathcal{A}_0}$, $\alpha, \beta \in \mathbb{Z}_{\geq 0}^{\mathcal{A} \setminus \mathcal{A}_0}$ and $\alpha', \beta' \in \mathbb{Z}_{\geq 0}^{\mathcal{A}' \setminus \mathcal{A}_0}$. By Lemma 2.5, there is an element $\gamma_0 \in \mathbb{Z}_{\geq 0}^{\mathcal{A}_0}$ such that $m_{\gamma_0} + m_\alpha$ and $m_{\gamma_0} + m_\beta$ belong to S_{σ_0} , so we can find $\alpha_1, \beta_1 \in \mathbb{Z}_{\geq 0}^{\mathcal{A}_0}$ such that

$$(\gamma_0 + \alpha) - \alpha_1, (\gamma_0 + \beta) - \beta_1 \in L_{\mathcal{A}_0}. \quad (3.4)$$

By (3.3) and (3.4), we get

$$(\alpha_0 + \alpha_1 + \alpha') - (\beta_0 + \beta_1 + \beta') \in L_{\mathcal{A}'},$$

and thus the binomial

$$\begin{aligned} x^{\gamma_0} (x^{\alpha_0 + \alpha + \alpha'} - x^{\beta_0 + \beta + \beta'}) \\ = \left[x^{\alpha_0 + \alpha'} (x^{\gamma_0 + \alpha} - x^{\alpha_1}) - x^{\beta_0 + \beta'} (x^{\gamma_0 + \beta} - x^{\beta_1}) \right] + \left[x^{\alpha_0 + \alpha_1 + \alpha'} - x^{\beta_0 + \beta_1 + \beta'} \right] \end{aligned}$$

belongs to $I_{\mathcal{A}} + I_{\mathcal{A}'}$. Then the prime ideal $I_{\mathcal{B}} / (I_{\mathcal{A}} + I_{\mathcal{A}'})$ defining $\mathbf{V}(I_{\mathcal{B}})$ is contained in the prime ideal $\iota^{-1}(0_{A_d})$ defining $\overline{\Gamma}_f \cap \text{Spec } A$. Therefore the claim follows from the dimension equality

$$\dim \overline{\Gamma}_f = \text{rank } N = \text{rank } M = \dim \mathbf{V}(I_{\mathcal{B}}).$$

According to

$$\text{Cone}(\mathcal{B})^\vee = (\sigma^\vee + (\sigma')^\vee)^\vee = \sigma \cap \sigma'$$

and Fact 2.4, it follows that $U_{\sigma \cap \sigma'}$ is the normalization of

$$\mathbf{V}(I_{\mathcal{B}}) \simeq \text{Spec}(\mathbb{C}[\mathbb{Z}_{\geq 0}^{\mathcal{B}}] / I_{\mathcal{B}}),$$

which proves (2).

In general, if $\mathcal{A} \cap \mathcal{A}' \subsetneq \mathcal{A}_0$, we can modify the above proof as follows: The fiber product $U_\sigma \times_{U_{\sigma_0}} U_{\sigma'}$ is the spectrum of the ring

$$A = \mathbb{C}[\mathbb{Z}_{\geq 0}^{\mathcal{A}}] / I_{\mathcal{A}} \otimes_{\mathbb{C}[\mathbb{Z}_{\geq 0}^{\mathcal{A}_0}] / I_{\mathcal{A}'}} \mathbb{C}[\mathbb{Z}_{\geq 0}^{\mathcal{A}'}] / I_{\mathcal{A}'} \simeq \mathbb{C}[\text{S}_{\text{split}}] / I_{\mathcal{A}} + I_{\mathcal{A}'}, \quad (3.5)$$

where S_{split} is the semigroup (2.2). Let L be the kernel of the map of lattices

$$\begin{aligned} \text{S}_{\text{split}} &\longrightarrow M \\ (\alpha_0, \alpha, \alpha') &\longmapsto m_{\alpha_0} + m_\alpha + m_{\alpha'}, \end{aligned}$$

and let I be the prime ideal of $\mathbb{C}[\text{S}_{\text{split}}]$ defined by

$$I = \langle x^\alpha - x^\beta \mid \alpha, \beta \in \text{S}_{\text{split}} \text{ and } \alpha - \beta \in L \rangle. \quad (3.6)$$

Then we claim that $\overline{\Gamma}_f \cap \text{Spec } A = V(I)$, and the same proof works for the general case. $\square$

The following theorem gives us a combinatorial criterion for the normality of graph closures.

Theorem 3.2 *Let f be as in Theorem 3.1. Then the following statements are equivalent:*

- (i) $\overline{\Gamma}_f$ is normal.
- (ii) $\overline{\Gamma}_f$ is the toric variety of the fan $\widetilde{\Sigma}$ defined in (2.6).
- (iii) For all (maximal cone) σ and σ' contained in some cone of Σ_0 ,

$$S_\sigma + S_{\sigma'} = S_{\sigma \cap \sigma'}. \quad (3.7)$$

Proof The equivalence (i) $\Leftrightarrow$ (ii) follows from Theorem 3.1. For (ii) $\Leftrightarrow$ (iii), we claim that (iii) is equivalent to saying that the natural morphism $X_{\widetilde{\Sigma}} \rightarrow X_\Sigma \times X_{\Sigma'}$, called ψ , is a closed immersion. Indeed, the morphism ψ is defined locally by

$$\begin{aligned} \mathbb{C}[S_\sigma] \otimes_{\mathbb{C}} \mathbb{C}[S_{\sigma'}] &\longrightarrow \mathbb{C}[S_{\sigma \cap \sigma'}] \\ \chi^m \otimes \chi^{m'} &\longmapsto \chi^{m+m'}, \end{aligned} \quad (3.8)$$

and it is a closed immersion if and only if (3.8) is surjective.

Since ψ is proper, we have the birational morphism

$$\psi: X_{\widetilde{\Sigma}} = \overline{T_N} \longrightarrow \overline{T_N \times_{T_N} T_N} = \overline{\Gamma}_f$$

and thus $\overline{\Gamma}_f$ is the image of ψ . Therefore the closed immersion $\psi: X_{\widetilde{\Sigma}} \rightarrow X_\Sigma \times X_{\Sigma'}$ gives the isomorphism $X_{\widetilde{\Sigma}} \xrightarrow{\sim} \overline{\Gamma}_f$. $\square$

As an application of Theorem 3.2, we are going to prove that $\overline{\Gamma}_f$ is a normal toric variety for certain toric flips f .

Let $f: X_\Sigma \dashrightarrow X_{\Sigma'}$ be a toric flip as in (2.11). Using the notation in Sect. 2.3, we assume for simplicity that $X_{\Sigma_0} = U_{\sigma_0}$.

Corollary 3.3 *Let $f: X_\Sigma \dashrightarrow X_{\Sigma'}$ and X_{Σ_0} be as above. Then $\overline{\Gamma}_f$ is normal if X_Σ satisfies the one of the following conditions.*

- (1) *Assume that $|J_+| = 2$, say $J_- \cup J_+ = \{1, 2, \dots, n-1\}$ and $J_+ = \{n, n+1\}$. The cone $\sigma_{\{1, \dots, n\}} \subseteq N_{\mathbb{R}} \simeq \mathbb{R}^n$ is smooth.*
- (2) *The toric variety X_Σ is smooth.*

Proof To simplify the notation, set $\sigma_i = \sigma_{\{1, \dots, n+1\} \setminus \{i\}}$. By Theorem 3.2, it suffices to check (3.7) holds when $(\sigma, \sigma') = (\sigma_j, \sigma_i)$ for $i \in J_-$ and $j \in J_+$. The inclusion $S_{\sigma \cap \sigma'} \supseteq S_\sigma + S_{\sigma'}$ follows directly from the general fact that $\sigma^\vee + (\sigma')^\vee = (\sigma \cap \sigma')^\vee$. Set

$$u := - \sum_{i \in J_-} b_i u_i = \sum_{j \in J_+} b_j u_j. \quad (3.9)$$

Then $\sigma \cap \sigma' = \text{Cone}(u_k, u \mid k \notin \{i, j\})$.

First assume that (1) holds. Without loss of generality, we may assume that $i = 1$. If $j = n+1$, then the cone $(\sigma \cap \sigma')^\vee$ is generated by dual vectors

$$\{u_1^\vee, u_n^\vee\} \cup \{-b_1 u_i^\vee + b_i u_1^\vee\}_{i \in J_- \setminus \{1\}} \cup \{u_j^\vee\}_{j \in J_0}.$$

Since the cone $\sigma_{\{u_1, \dots, u_n\}}$ is smooth, we see that $\{u_1^\vee, \dots, u_n^\vee\}$ forms a $\mathbb{Z}$ -basis of M . Hence

$$\begin{aligned} S_{\sigma \cap \sigma'} &= \text{Cone}(u_1^\vee, -b_1 u_i^\vee + b_i u_1^\vee, u_j^\vee \mid i \in J_- \setminus \{1\}, j \in J_0) \cap M + \text{Cone}(u_n^\vee) \cap M \\ &\subseteq S_{\sigma'} + S_\sigma. \end{aligned}$$

If $j = n$, by the linear relation (3.9), we have

$$\overline{(\sigma \cap \sigma')^\vee \setminus \sigma'^\vee} = \text{Cone}(u_2, \dots, u_{n-1}, u_{n+1}, u, -u_n)^\vee = \text{Cone}(u_2, \dots, u_{n-1}, u, -u_n)^\vee,$$

which is generated by dual vectors

$$u_1^\vee, \dots, u_{n-2}^\vee, -b_1 u_{n-1}^\vee + b_{n-1} u_1^\vee, -u_n^\vee$$

Using that $\{u_1^\vee, \dots, u_n^\vee\}$ is a $\mathbb{Z}$ -basis of M again, we get

$$\overline{(\sigma \cap \sigma')^\vee \setminus \sigma'^\vee} \cap M \subseteq S_{\sigma'} + S_\sigma.$$

According to

$$(\sigma \cap \sigma')^\vee = \overline{(\sigma \cap \sigma')^\vee \setminus \sigma'^\vee} \cup (\sigma')^\vee,$$

it follows that $S_{\sigma \cap \sigma'} \subseteq S_\sigma + S_{\sigma'}$, which proves Corollary 3.3 in the situation (1).

Now assume that (2) holds. For simplicity, we further assume that $1 \in J_-$, $n+1 \in J_+$ and $(\sigma, \sigma') = (\sigma_{n+1}, \sigma_1)$. For $m \in S_{\sigma \cap \sigma'}$, we let

$$m_1 = \sum_{j \in J_0 \cup J_+} \langle m, u_j \rangle u_j^\vee \quad \text{and} \quad m_2 = \sum_{i \in J_-} \langle m, u_i \rangle u_i^\vee.$$

According to $(\sigma \cap \sigma')^\vee = \text{Cone}(u_k, u \mid k \notin \{1, n+1\})^\vee$, it follows that $m_1 \in \sigma^\vee$ and $m_2 \in (\sigma')^\vee$. Hence $m = m_1 + m_2 \in S_\sigma + S_{\sigma'}$, as required. $\square$

Remark 3.4 While the condition 1 in Corollary 3.3 provides a criterion for general dimensions, the 3-dimensional case (characterized by $|J_{\pm}| = 2$ and $|J_0| = 0$) admits a more refined description. Specifically, for general dimensions, condition (1) requires at least one maximal cone $\sigma_J \in \Sigma$ (where J contains J_-) to be smooth. However, since the normality of $\bar{\Gamma}_f$ is symmetric with respect to X_{Σ} and $X_{\Sigma'}$, the same conclusion holds if a maximal cone $\sigma_J \in \Sigma'$ (where J contains J_+) is smooth. Combining these observations, we can strengthen the condition in 3-dimensional as follows: $\bar{\Gamma}_f$ is normal provided that the cone σ_J is smooth for some $J \subseteq \{1, 2, 3, 4\}$ with $|J| = 3$.

We provide two counterexamples to illustrate the necessity of the two conditions in Corollary 3.3 (1). These conditions require the smoothness of at least one maximal cone and that $|J_+| = 2$. In each example, normality fails when exactly one of these requirements is violated.

Example 3.5 Consider four integer vectors in $N = \mathbb{Z}^3$

$$u_1 = (1, 0, 0), \quad u_2 = (1, 3, 0), \quad u_3 = (0, 1, 2), \quad u_4 = (2, 1, -3),$$

which satisfy the wall relation

$$-7u_1 - 5u_2 + 9u_3 + 6u_4 = 0.$$

Then we have $J_- = \{1, 2\}$, $J_+ = \{3, 4\}$, $J_0 = \emptyset$, and two simplicial fans Σ and Σ' defined as in Sect. 2.2. Note that σ_J is not smooth for any $J \subseteq \{1, 2, 3, 4\}$ with $|J| = 3$. Moreover, the graph closure of $f: X_{\Sigma} \dashrightarrow X_{\Sigma'}$ is not normal.

Indeed, we let cones $\sigma = \sigma_4$, $\sigma' = \sigma_1$, where $\sigma_i = \sigma_{\{1, \dots, 4\} \setminus \{i\}}$. We find that $S_{\sigma} + S_{\sigma'}$ has a generating set consisting of

$$\begin{pmatrix} 6 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -2 \\ 2 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ -2 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ -3 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 3 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 4 \\ -2 \end{pmatrix}.$$

However, the lattice point $(-1, 1, 0) \in S_{\sigma \cap \sigma'}$ but not in $S_{\sigma} + S_{\sigma'}$. Therefore (3.7) does not hold for (σ, σ') .

This counterexample generalizes straightforwardly to higher dimensions. For $n > 3$, consider the embedding $\phi: \mathbb{Z}^3 \hookrightarrow N = \mathbb{Z}^n$ defined by mapping $\mathbb{Z}^3$ into the first three components of $\mathbb{Z}^n$. Consider the wall relation in N

$$-7u_1 - 5u_2 + \sum_{j=3}^{n-1} 0 \cdot u_j + 9u_n + 6u_{n+1} = 0,$$

where

$$u_1 = \phi(1, 0, 0), \quad u_2 = \phi(1, 3, 0), \quad u_n = \phi(0, 1, 2), \quad u_4 = \phi(2, 1, -3)$$

and $\{u_j\}_{j=3}^{n-1}$ are the remaining standard basis vector $\{e_4, \dots, e_n\}$. By a similar argument to the 3-dimensional case, it follows that $\bar{\Gamma}_f$ is not normal.

Example 3.6 Let $\mathcal{S}_0 := \{u_i \mid 1 \leq i \leq 5\}$ be a $\mathbb{Z}$ -basis of $N \simeq \mathbb{Z}^5$, and let $\sigma_0 = \text{Cone}(\mathcal{S}_0)$. Consider the affine toric variety $X_{\Sigma_0} = U_{\sigma_0}$ and the wall relation

$$-3u_1 - 2u_2 - u_3 + 3u_4 + 2u_5 + u_6 = 0.$$

Then we have $J_- = \{1, 2, 3\}$, $J_+ = \{4, 5, 6\}$, $J_0 = \emptyset$, and two simplicial fans Σ and Σ' defined as in Sect. 2.2. Although the cone $\sigma_{J_- \cup \{4, 5\}}$ is smooth by construction, the graph closure of $f: X_{\Sigma} \dashrightarrow X_{\Sigma'}$ is not normal.

Indeed, we let cones $\sigma = \sigma_4$, $\sigma' = \sigma_1$, where $\sigma_i = \sigma_{\{1, \dots, 6\} \setminus \{i\}}$. We find that $S_\sigma + S_{\sigma'}$ has a generating set consisting of

$$\begin{pmatrix} 2 \\ 0 \\ 0 \\ 0 \\ 0 \\ 3 \end{pmatrix} \quad \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 0 \\ 0 \\ 2 \\ 0 \\ 0 \\ 1 \end{pmatrix} \quad \begin{pmatrix} -1 \\ 0 \\ 3 \\ 0 \\ 0 \\ 0 \end{pmatrix} \\ \begin{pmatrix} -1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 0 \\ 3 \\ 0 \\ 2 \\ 0 \\ 0 \end{pmatrix} \quad \begin{pmatrix} -2 \\ 3 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 0 \\ 0 \\ 1 \\ -1 \\ 2 \\ 0 \end{pmatrix} \\ \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \quad \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \\ \begin{pmatrix} 0 \\ 0 \\ 0 \\ -1 \\ 1 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 0 \\ 0 \\ 0 \\ -1 \\ 0 \\ 0 \end{pmatrix} \quad \begin{pmatrix} -1 \\ 2 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

However, the lattice point $(-1, 2, 0, -1, 2) \in S_{\sigma \cap \sigma'}$ but not in $S_\sigma + S_{\sigma'}$. Therefore (3.7) does not hold for (σ, σ') .

4 Fiber products

In this section, we study the fiber product $X := X_\Sigma \times_{X_{\Sigma_0}} X_{\Sigma'}$ in the category of schemes for two refinements Σ and Σ' of a fan Σ_0 . In particular, we will concentrate on toric flips.

Proposition 4.1 *Let $f: X_\Sigma \dashrightarrow X_{\Sigma'}$ be the birational map induced by toric morphisms $X_\Sigma \rightarrow X_{\Sigma_0} \leftarrow X_{\Sigma'}$. Then the following statements are equivalent:*

- (1) *The fiber product X is a normal variety.*
- (2) *The scheme X is irreducible, reduced and the graph closure $\overline{\Gamma}_f$ is a normal toric variety.*
- (3) *$X = X_{\widetilde{\Sigma}}$.*

Note that the conditions in (2) are all local properties by Theorem 3.2.

Proof Note that X_Σ is irreducible and so is the graph closure $\overline{T_N \times_{T_N} T_N} = \overline{\Gamma}_f$. If the fiber product X satisfies one of the conditions (1), (2), (3), then X is irreducible and reduced and thus $\overline{\Gamma}_f = X = X_{\text{red}}$. Therefore Proposition 4.1 follows from Theorem 3.1. $\square$

Example 4.2 In this example, we will illustrate how Proposition 4.1 provides a method to construct a non-normal fiber product. Let $\sigma_0 = \text{Cone}(u_1, \dots, u_5)$ be a strongly convex cone in $N \simeq \mathbb{Z}^3$ such that $\text{Cone}(u_i, u_j)$ is a 2-dimensional face if and only if $|i - j| = 1$ or 4. To simplify the notation, let

$$\sigma_J = \text{Cone}(u_j \mid j \in J) \text{ for } J \subseteq \{1, 2, 3, 4, 5\}.$$

We start with the 3-dimensional toric variety X_Σ such that Σ contains three cones $\sigma_{\{1,2,4\}}$, $\sigma_{\{2,3,4\}}$, $\sigma_{\{1,4,5\}}$. If we perform two consecutive toric flips along the walls $\sigma_{\{2,4\}}$ and $\sigma_{\{1,4\}}$, then we will obtain the toric variety $X_{\Sigma'}$. Let Σ_0 be the fan removing the walls $\sigma_{\{2,4\}}$ and $\sigma_{\{1,4\}}$ from Σ . Then X_Σ and $X_{\Sigma'}$ are two varieties over X_{Σ_0} . We claim that the fiber product $X = X_\Sigma \times_{X_{\Sigma_0}} X_{\Sigma'}$ is not a normal toric variety.

According to Proposition 4.1, if X is a normal toric variety, then $X = X_{\widetilde{\Sigma}}$, where $\widetilde{\Sigma}$ is defined in (2.6). Let ϕ, ϕ' , and $\widetilde{\phi}$ be the birational morphisms from $X_\Sigma, X_{\Sigma'}$ and $X_{\widetilde{\Sigma}}$ to X_{Σ_0} respectively. Observe that the exceptional locus Z of ϕ is

$$\text{Exc}(\phi) = V(\sigma_{\{1,4\}}) \cup V(\sigma_{\{2,4\}}) = \mathbb{P}^1 \cup_{\text{pt}} \mathbb{P}^1,$$

which maps to $S := V(\sigma_0)$. Similarly, the exceptional locus Z' of ϕ' is $\mathbb{P}^1 \cup_{\text{pt}} \mathbb{P}^1$. Then

$$(\mathbb{P}^1 \cup_{\text{pt}} \mathbb{P}^1) \times (\mathbb{P}^1 \cup_{\text{pt}} \mathbb{P}^1) \subseteq X = X_{\widetilde{\Sigma}} \quad (4.1)$$

contained in $\text{Exc}(\widetilde{\phi})$. Since $\text{Exc}(\widetilde{\phi})$ is the union of three $\mathbb{P}^1 \times \mathbb{P}^1$, it can be contained in at most three disjoint $(\mathbb{C}^\times)^2$. However, (4.1) indicates that $\text{Exc}(\widetilde{\phi})$ contains at least four disjoint $(\mathbb{C}^\times)^2$, which leads to a contradiction. So we conclude that X is not a normal toric variety.

4.1 Irreducibility

We are going to prove that the reduced scheme associated to the fiber product under a toric flip is a toric variety.

Theorem 4.3 *Let $f: X_\Sigma \dashrightarrow X_{\Sigma'}$ be a toric flip via toric morphisms $X_\Sigma \rightarrow X_{\Sigma_0} \leftarrow X_{\Sigma'}$ (2.10) and $X = X_\Sigma \times_{X_{\Sigma_0}} X_{\Sigma'}$ be the fiber product. Then we have:*

- (1) *The reduced scheme X_{red} associated to X is the toric variety $\overline{\Gamma}_f$.*
- (2) *The normalization of X_{red} is the toric variety $X_{\widehat{\Sigma}}$ defined by the fan (2.6).*

Proof If X_{red} is irreducible, then Theorem 4.3 follows from Theorem 3.1 and $\overline{\Gamma}_f = X_{\text{red}}$. As this is a local question, we may assume that X_{Σ_0} is an affine toric variety U_{σ_0} defined by the cone $\sigma_0 = \text{Cone}(u_1, \dots, u_{n+1})$ and there is a wall relation (2.8)

$$u := - \sum_{i \in J_-} b_i u_i = \sum_{j \in J_+} b_j u_j. \quad (4.2)$$

Without loss of generality, we can assume that $1 \in J_-$ and $n+1 \in J_+$. Set

$$\sigma = \sigma_{\{1, \dots, n+1\} \setminus \{n+1\}} \text{ and } \sigma' = \sigma_{\{1, \dots, n+1\} \setminus \{1\}}.$$

To prove (1), it suffices to show that the reduced scheme associated to $U_\sigma \times_{U_{\sigma_0}} U_{\sigma'}$ is an affine toric variety.

Let $\mathcal{A}_0, \mathcal{A}$ and $\mathcal{A}' \subseteq M$ be generating sets of S_{σ_0}, S_σ and $S_{\sigma'}$ respectively. We may assume that $\mathcal{A} \cap \mathcal{A}' = \mathcal{A}_0$ because of the inclusion $\sigma_0^\vee \subseteq \sigma^\vee \cap (\sigma')^\vee$. Set $\mathcal{B} = \mathcal{A} \cup \mathcal{A}'$. Clearly, the fiber product $U_\sigma \times_{U_{\sigma_0}} U_{\sigma'}$ is the spectrum of the ring (3.1) as we have seen in the proof of Theorem 3.1. Then the irreducibility of X_{red} follows from the claim that

$$\sqrt{I_{\mathcal{A}} + I_{\mathcal{A}'}} = I_{\mathcal{B}}. \quad (4.3)$$

To see (4.3), the inclusion $\sqrt{I_{\mathcal{A}} + I_{\mathcal{A}'}} \subseteq I_{\mathcal{B}}$ follows from the fact that $I_{\mathcal{B}}$ is a prime ideal and $I_{\mathcal{A}} + I_{\mathcal{A}'} \subseteq I_{\mathcal{B}}$. For the other inclusion, the proof will be divided into four steps.

Step 1. As in the proof of Theorem 3.1, we pick

$$(\alpha_{01} + \alpha_1 + \alpha'_1) - (\alpha_{02} + \alpha_2 + \alpha'_2) \in L_{\mathcal{B}} \quad (4.4)$$

where $\alpha_{01}, \alpha_{02} \in \mathbb{Z}_{\geq 0}^{\mathcal{A}_0}$, $\alpha_1, \alpha_2 \in \mathbb{Z}_{\geq 0}^{\mathcal{A} \setminus \mathcal{A}_0}$ and $\alpha'_1, \alpha'_2 \in \mathbb{Z}_{\geq 0}^{\mathcal{A}' \setminus \mathcal{A}_0}$, using the identification (3.2). To simplify notation, we set $\beta_1 = \alpha_{01} + \alpha_1 + \alpha'_1$ and $\beta_2 = \alpha_{02} + \alpha_2 + \alpha'_2$. Suppose that there exists $k_i \in \mathbb{N}$ such that

$$x^{k_i \alpha_{0i}} (x^{\beta_1} - x^{\beta_2}) \in \sqrt{I_{\mathcal{A}} + I_{\mathcal{A}'}} \quad (4.5)$$

for $i = 1, 2$. Using binomial expansion

$$(x^{\beta_1} - x^{\beta_2})^{k_1+k_2+1} = \sum_{k=0}^{k_1+k_2} \binom{k_1+k_2}{k} x^{k\beta_1} x^{(k_1+k_2-k)\beta_2} (x^{\beta_1} - x^{\beta_2})$$

and that either $k \geq k_1$ or $k_1 + k_2 - k \geq k_2$ holds, we get $x^{\beta_1} - x^{\beta_2} \in \sqrt{I_{\mathcal{A}} + I_{\mathcal{A}'}}$ and thus verify (4.3), as required.

To get (4.5), we can replace, if necessary, (4.4) by the following two relations

$$(k_1 + 1)\alpha_{01} + \alpha_1 + \alpha'_1 \equiv (k_1\alpha_{01} + \alpha_{02}) + \alpha_2 + \alpha'_2 \pmod{L_{\mathcal{B}}}, \quad (4.6)$$

$$(k_2\alpha_{02} + \alpha_{01}) + \alpha_1 + \alpha'_1 \equiv (k_2 + 1)\alpha_{02} + \alpha_2 + \alpha'_2 \pmod{L_{\mathcal{B}}}. \quad (4.7)$$

In general, the pair $(\alpha_{01}, \alpha_{02})$ can be replaced by any pair in $\{\alpha_{01}, \alpha_1, \alpha'_1\} \times \{\alpha_{02}, \alpha_2, \alpha'_2\}$. This process of replacements will occur frequently in the following algorithm of finding such pair (k_1, k_2) of natural numbers.

Step 2. To simplify notation, we set $m_{0i} = m_{\alpha_{0i}}$, $m_i = m_{\alpha_i}$ and $m'_i = m_{\alpha'_i}$ for $i = 1, 2$. If $m_{01}, m_{02} \in \sigma_0^\vee \setminus \text{Cone}(u_1)^\perp$, then we can take $k_1, k_2 \in \mathbb{N}$ such that $m_{0i} + (1/k_i)m'_j \in \sigma_0^\vee$ for each $j = 1, 2$. Set $m_{ij} = k_i m_{0i} + m'_j$ for $1 \leq i, j \leq 2$. We find that $m_{ij} \in \sigma_0^\vee \cap M = S_{\sigma_0}$ and thus $m_{ij} = m_{\beta_{ij}}$ for some $\beta_{ij} \in \mathbb{Z}_{\geq 0}^{\mathcal{A}_0}$. Then the relation (4.6) reduces to

$$\begin{aligned} k_1 \alpha_{01} + \alpha'_1 &\equiv \beta_{11} \pmod{L_{\mathcal{A}'}} \\ \beta_{12} &\equiv k_1 \alpha_{01} + \alpha'_2 \pmod{L_{\mathcal{A}'}} \\ \beta_{11} + \alpha_{01} + \alpha_1 &\equiv \beta_{12} + \alpha_{02} + \alpha_2 \pmod{L_{\mathcal{A}}}. \end{aligned}$$

Hence, the binomial corresponding to the relation (4.6) is in $I_{\mathcal{A}} + I_{\mathcal{A}'}$. A similar argument holds for (4.7).

We also can replace the condition $\sigma_0^\vee \setminus \text{Cone}(u_1)^\perp$ with $\sigma_0^\vee \setminus \text{Cone}(u_{n+1})^\perp$. Then we can take $k_1, k_2 \in \mathbb{N}$ such that $m_{0i} + (1/k_i)m_j \in \sigma_0^\vee$ for each $j = 1, 2$, and the similar argument also works.

Step 3. We use the following notation for the remainder of the proof. Given a face τ of a cone σ , we define $\tau^* = \sigma^\vee \cap \tau^\perp$, the dual face of τ . Then τ^* is a face of $\sigma^\vee$.

Consider the face $\sigma_{J_0 \cup J_-}^*$ of $\sigma^\vee$ and the face $\sigma_{J_0 \cup J_+}^*$ of $(\sigma')^\vee$. In the following cases, we can immediately check that:

- If $m_{0i} \neq 0$, then $k_i m_{0i} \in [\sigma_0^\vee \setminus \text{Cone}(u_1)^\perp] + \sigma_{J_0 \cup J_+}^*$ for some $k_i \in \mathbb{N}$.
- If $m_i \notin \sigma_{J_0 \cup J_-}^*$, then $k_i m_i \in [\sigma_0^\vee \setminus \text{Cone}(u_{n+1})^\perp] + \sigma_{J_0 \cup J_-}^*$ for some $k_i \in \mathbb{N}$.
- If $m'_i \notin \sigma_{J_0 \cup J_+}^*$, then $k_i m'_i \in [\sigma_0^\vee \setminus \text{Cone}(u_1)^\perp] + \sigma_{J_0 \cup J_+}^*$ for some $k_i \in \mathbb{N}$.

We are going to verify (4.5) in the above cases. Suppose that one of the above three conditions holds for $i = 1, 2$, and let γ_i denote the vector corresponding to the satisfied lattice point, where $\gamma_i \in \{\alpha_{0i}, \alpha_i, \alpha'_i\}$. Replace (4.4) with (4.6) and (4.7) on the pairs (γ_1, γ_2) and (k_1, k_2) as in Step 1. By replacing $k_i \gamma_i$ with the above expression on the both sides of (4.6) and (4.7), we can reduce it to Step 2.

For example, we treat the case $\gamma_1 = \alpha_{01}$ and $\gamma_2 = \alpha_{02}$, which is one of the nine cases. We take

$$m_{\gamma_{0i}} \in [\sigma_0^\vee \setminus \text{Cone}(u_1)^\perp] \quad \text{and} \quad m_{\gamma'_i} \in \sigma_{J_0 \cup J_+}^*$$

for some $\gamma_{0i} \in \mathbb{Z}_{\geq 0}^{\mathcal{A}_0}$, $\gamma'_i \in \mathbb{Z}_{\geq 0}^{\mathcal{A}' \setminus \mathcal{A}_0}$ such that $k_i m_{0i} = m_{\gamma_{0i}} + m_{\gamma'_i}$. Then the relation (4.6) reduces to the congruences $k_1 \alpha_{01} \equiv \gamma_{01} + \gamma'_1 \pmod{L_{\mathcal{A}'}}$ and

$$(\gamma_{01} + \alpha_{01}) + \alpha_1 + (\alpha'_1 + \gamma'_1) \equiv (\gamma_{01} + \alpha_{02}) + \alpha_{02} + (\alpha'_2 + \gamma'_1) \pmod{L_{\mathcal{B}}}. \quad (4.8)$$

Note that $m_{\gamma_{01} + \alpha_{01}}, m_{\gamma_{01} + \gamma_{02}} \in \sigma_0^\vee \setminus \text{Cone}(u_1)^\perp$, since $m_{\gamma_{01}} \in \sigma_0^\vee \setminus \text{Cone}(u_1)^\perp$. So we can apply Step 2 to show that the binomial corresponding to the relation 4.8 is in $\sqrt{I_{\mathcal{A}} + I_{\mathcal{A}'}}$, and thus the binomial corresponding to the relation (4.6) is in $\sqrt{I_{\mathcal{A}} + I_{\mathcal{A}'}}$. A similar argument holds for (4.7).

Step 4. The remaining case to consider is when at least one of $i \in \{1, 2\}$ does not satisfy the condition in Step 3. Without loss of generality, we assume that $i = 2$, that is,

$$m_{02} = 0, \quad m_2 \in \sigma_{J_0 \cup J_-}^*, \quad m'_2 \in \sigma_{J_0 \cup J_+}^*.$$

Recall that u is the vector (4.2) associated to the wall relation and thus $\text{Cone}(u)$ is a face of $\sigma \cap \sigma'$. Since

$$\sigma_{J_0 \cup J_-}^* + \sigma_{J_0 \cup J_+}^* = (\sigma \cap \sigma')^\vee \cap \text{Cone}(u)^\perp \quad (4.9)$$

is a face of $(\sigma \cap \sigma')^\vee$ and $m_{01} + m_1 + m'_1 = m_2 + m'_2$ belongs to the left hand side of (4.9), we get

$$m_{01} \in S_{\sigma_0} \cap (\sigma_{J_0 \cup J_-}^* + \sigma_{J_0 \cup J_+}^*) = \{0\}, \quad m_1 \in \sigma_{J_0 \cup J_-}^*, \quad m'_1 \in \sigma_{J_0 \cup J_+}^*.$$

Therefore we find that

$$m_1 - m_2 = m'_1 - m'_2 \in \sigma_{J_0 \cup J_-}^\perp \cap \sigma_{J_0 \cup J_+}^\perp = \{0\}.$$

Since $\sigma_0^\vee$ is strongly convex, it forces $\alpha_{01} = \alpha_{02} = 0$, and thus

$$\begin{aligned} \alpha_1 &\equiv \alpha_2 \pmod{L_{\mathcal{A}}}, \\ \alpha_{01} + \alpha'_1 &\equiv \alpha_{02} + \alpha'_2 \pmod{L_{\mathcal{A}'}}. \end{aligned}$$

Hence the binomial

$$x^{\beta_1} - x^{\beta_2} = x^{\alpha_{01} + \alpha'_1} (x^{\alpha_1} - x^{\alpha_2}) + x^{\alpha_2} (x^{\alpha_{01} + \alpha'_1} - x^{\alpha_{02} + \alpha'_2})$$

belongs to $I_{\mathcal{A}} + I_{\mathcal{A}'}$, as required.

The second statement follows from Theorem 3.1. $\square$

Remark 4.4 For any (not necessarily maximal) cone $\sigma \in \Sigma$ and $\sigma' \in \Sigma'$ contained in a cone $\sigma_0 \in \Sigma_0$, we also have the equality

$$\sqrt{I_{\mathcal{A}} + I_{\mathcal{A}'}} = I \quad (4.10)$$

where I is the ideal of $\mathbb{C}[\text{Split}]$ as defined in (3.6) and $I_{\mathcal{A}}$ and $I_{\mathcal{A}'}$ are corresponding toric ideals of S_σ and $S_{\sigma'}$ respectively. Indeed, by Theorem 3.1 and Proposition 4.1, we have

$$\text{Spec}(\mathbb{C}[\text{Split}]/\sqrt{I_{\mathcal{A}} + I_{\mathcal{A}'}}) = (U_\sigma \times_{U_{\sigma_0}} U_{\sigma'})_{\text{red}} = \overline{\Gamma}_f \cap (U_\sigma \times_{U_{\sigma_0}} U_{\sigma'}) = \text{Spec}(\mathbb{C}[\text{Split}]/I)$$

Note that I is equal to the toric ideal $I_{\mathcal{A} \cup \mathcal{A}'}$ if $\sigma_0 = \sigma + \sigma'$.

4.2 Reduced property

In this subsection, we study the reduced property of X . When we concentrate on a 3-dimensional toric flip f satisfied the condition (1) in Corollary 3.3, by Corollary 3.3, Proposition 4.1, and Theorem 4.3, the following statements are equivalent:

- (i) $X = X_\Sigma \times_{X_{\Sigma_0}} X_{\Sigma'}$ is a normal toric variety.
- (ii) $X = X_{\widetilde{\Sigma}}$,
- (iii) X is reduced.

Since the property of being reduced is local, we may assume X_{Σ_0} is an affine toric variety U_{σ_0} defined by $\sigma_0 = \text{Cone}(u_1, u_2, u_3, u_4)$, and $X_\Sigma, X_{\Sigma'}$ are defined by the wall relation (2.8) as in Sect. 2.2. To simplify notation, let $\sigma_i = \sigma_{\{1,2,3,4\} \setminus \{i\}}$ and $U_{ji} := U_{\sigma_j} \times_{U_{\sigma_0}} U_{\sigma_i}$.

We normalize our wall relation such that $b_i \in \mathbb{Z}$ and $\gcd(b_1, b_2, b_3, b_4) = 1$. Note that since σ_4 is a smooth cone, $\{u_1, u_2, u_3\}$ is a $\mathbb{Z}$ -basis of N . From the wall relation (2.8), we have that $b_4 \mid b_i$ for $i = 1, 2, 3$, and thus $b_4 = 1$.

First, we show a numerical criterion for the reduced property on the affine piece U_{31} .

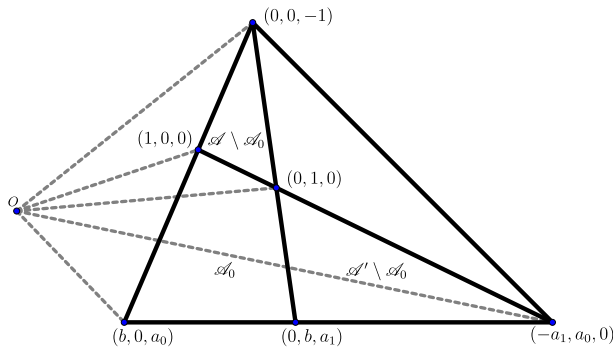


Fig. 1 Dual cones of σ and σ'

Lemma 4.5 Let $\{a\}_b$ denote the remainder of a divided by b . If $g = \gcd(b_1, b_2) > 0$ and $b_i = -gb'_i$ for $i = 1, 2$, then the following statements are equivalent:

- (iv) U_{31} is reduced.
- (v) For all $0 \leq \lambda \leq b'_1 b'_2$, there exists a non-negative integer $y \leq \lambda/b'_1$ such that

$$\{g\lambda\}_{b_3} \geq g \cdot \{\lambda - b'_1 y\}_{b'_2}. \quad (4.11)$$

Proof First of all, we have to determine the dual cones of $\sigma = \sigma_3$ and $\sigma' = \sigma_1$, as illustrated in Fig. 1, where we identify $M \simeq \mathbb{Z}^3$ by the dual basis $\{u_1^\vee, u_2^\vee, u_3^\vee\}$ of $\{u_1, u_2, u_3\}$. Note that the coordinates in Fig. 1 only represent points on that ray, not that these six points lie in the same plane.

We use the same notation in the proof of Theorem 4.3. Observe that (iv): U_{31} is reduced if and only if (iv'): $x^{\alpha_0 + \alpha + \alpha'} - x^{\beta_0 + \beta + \beta'} \in I_{\mathcal{A}} + I_{\mathcal{A}'}$ when $(\alpha_0 + \alpha + \alpha') - (\beta_0 + \beta + \beta') \in L_{\mathcal{B}}$.

The **key point** is that the generating set $\mathcal{A}$ can be selected to be $\mathcal{A}_0 \cup \{-u_3^\vee\}$. Let $\gamma \in \mathbb{Z}_{\geq 0}^{\mathcal{A} \setminus \mathcal{A}_0}$ such that $-u_3^\vee = m_\gamma$. Notice that γ is the unique vector in $\mathbb{Z}_{\geq 0}^{\mathcal{A}}$ such that $m_\gamma = -u_3^\vee$.

Define two semi-groups

$$\Gamma = M \cap \overline{\sigma^\vee \setminus \sigma^\vee}, \quad \Gamma' = \Gamma \cap \text{Cone}(u_3)^\perp.$$

We **claim** that (iv') holds if and only if (iv''): $\Gamma \setminus \Gamma' \subseteq (S_{\sigma_0} \setminus 0) + \Gamma'$. Assume that (iv'') holds. For any $m \in \Gamma \setminus \Gamma'$, it is clear that $m - u_3^\vee \in S_{\sigma'}$ since

$$\langle m - u_3^\vee, u_3 \rangle = \langle m, u_3 \rangle - 1 \geq 0.$$

Take a lifting $\alpha_0, \beta_0 \in \mathbb{Z}_{\geq 0}^{\mathcal{A}_0}$, $\alpha', \beta' \in \mathbb{Z}_{\geq 0}^{\mathcal{A}' \setminus \mathcal{A}_0}$ such that $m = m_{\alpha_0 + \alpha'}$ and $m - u_3^\vee = m_{\beta_0 + \beta'}$. Then $x^{\alpha_0 + \gamma + \alpha'} - x^{\beta_0 + \beta'} \in I_{\mathcal{A}} + I_{\mathcal{A}'}$ since $(\alpha_0 + \gamma + \alpha') - (\beta_0 + \beta') \in L_{\mathcal{B}}$. By lemma 2.6, there exists a sequence $\{(\gamma_{0i}, \gamma_i, \gamma'_i)\}_{i=1}^m \subseteq S_{\text{split}}$ such that

$$(\gamma_{01}, \gamma_1, \gamma'_1) = (\alpha_0, \gamma, \alpha') \quad \text{and} \quad (\gamma_{0m}, \gamma_m, \gamma'_m) = (\beta_0, 0, \beta'),$$

and for each $1 \leq i \leq m-1$, either (2.4) or (2.5) holds. If i is the smallest index such that $\gamma_{i+1} \neq \gamma$, then (2.5) must hold for i , that is,

$$(\gamma_{0i} + \gamma_i) - (\gamma_{0(i+1)} + \gamma_{i+1}) \in L_{\mathcal{A}}.$$

If $m_{\gamma_{0i}} = 0$, then $m_{\gamma_{0(i+1)}} + m_{\gamma_{i+1}} = m_{\gamma_i} = -u_3^\vee$ in the face $\sigma_{\{12\}}^*$ of $\sigma^\vee$. This implies that $m_{\gamma_{0(i+1)}} \in \sigma_0^\vee \cap \sigma_{12}^* = \{0\}$ and $m_{\gamma_{i+1}} = -u_3^\vee$, and thus $\gamma_{i+1} = \gamma$, a contradiction. Hence

$m_{\gamma_{\sigma_i}} \neq 0$. Since $m_{\gamma_{0i}} + m_{\gamma_i} + m_{\gamma'_i}$ is a constant, we have

$$m_{\alpha_0} + m_{\gamma} + m_{\alpha'} = m_{\gamma_{0i}} + m_{\gamma_i} + m_{\gamma'_i} \implies m = m_{\alpha_0 + \alpha'} = m_{\gamma_{0i}} + m_{\gamma'_i} \in (S_{\sigma_0} \setminus 0) + \Gamma.$$

Furthermore, let $m_1 = m \in \Gamma \setminus \Gamma'$. Since $\Gamma \setminus \Gamma' \subseteq (S_{\sigma_0} \setminus 0) + \Gamma$, there exists $m_2 \in \Gamma$ such that $m_1 - m_2 \in (S_{\sigma_0} \setminus 0)$. By induction, if $m_i \in \Gamma \setminus \Gamma'$, then there exists $m_{i+1} \in \Gamma$ such that $m_i - m_{i+1} \in (S_{\sigma} \setminus 0)$. This process can be continuously done until m_{i+1} no longer lies in $\Gamma \setminus \Gamma'$. Note that this process will terminate, since $(\sigma')^\vee$ is strongly convex. Hence (iv'') holds. Conversely, we assume that (iv'') holds. It is clear that (iv'') is equivalent to

$$\Gamma \subseteq S_{\sigma_0} + \Gamma'. \quad (4.12)$$

For any $\alpha' \in \mathbb{Z}_{\geq 0}^{\mathcal{A}' \setminus \mathcal{A}_0}$ such that $m_{\alpha'} \in \Gamma$, we can decompose $m_{\alpha'} = m_{\beta_0} + m_{\beta'}$ by (4.12) and get

$$x^{\alpha'} - x^{\beta_0 + \beta'} \in I_{\mathcal{A}'}, \quad (4.13)$$

where $m_{\beta_0} \in S_{\sigma_0}$, $m_{\beta'} \in \Gamma'$ for some $\beta_0 \in \mathbb{Z}_{\geq 0}^{\mathcal{A}_0}$ and $\beta' \in \mathbb{Z}_{\geq 0}^{\mathcal{A}'}$. Since

$$0 < \langle m_{\alpha'}, u_3 \rangle = \langle m_{\beta_0}, u_3 \rangle,$$

we still have $m_{\beta_0} - u_3^\vee \in S_{\sigma_0}$ and $m_{\alpha'} - u_3^\vee \in S_{\sigma'}$. Take a lifting $\tilde{\beta}_0 \in \mathbb{Z}_{\geq 0}^{\mathcal{A}_0}$, $\alpha'' \in \mathbb{Z}_{\geq 0}^{\mathcal{A}'}$ such that $m_{\tilde{\beta}_0} = m_{\beta_0} - u_3^\vee$ and $m_{\alpha''} = m_{\alpha'} - u_3^\vee$. Then $m_{\alpha''} = m_{\tilde{\beta}_0} + m_{\beta'}$, and thus

$$x^{\beta_0 + \gamma} - x^{\tilde{\beta}_0} \in I_{\mathcal{A}}, \quad x^{\alpha''} - x^{\tilde{\beta}_0 + \beta'} \in I_{\mathcal{A}'}. \quad (4.14)$$

From (4.13) and (4.14), we conclude

$$x^{\gamma + \alpha'} - x^{\alpha''} \in I_{\mathcal{A}} + I_{\mathcal{A}'}. \quad (4.15)$$

Now, suppose that (iv') is false, say, there exists a relation

$$(\alpha_0 + k_1\gamma + \alpha') - (\beta_0 + k_2\gamma + \beta') \in L_{\mathcal{B}}$$

such that $x^{\alpha_0 + k_1\gamma + \alpha'} - x^{\beta_0 + k_2\gamma + \beta'} \notin I_{\mathcal{A}} + I_{\mathcal{A}'}$ for some $\alpha_0, \beta_0 \in \mathbb{Z}_{\geq 0}^{\mathcal{A}_0}$, $\alpha', \beta' \in \mathbb{Z}_{\geq 0}^{\mathcal{A}'}$ and $k_1, k_2 \in \mathbb{Z}_{\geq 0}$. We may assume that $k_2 = 0$ by eliminating $\min\{k_1, k_2\}\gamma$, and $k_1 \geq 0$ is the smallest integer such that the above relation holds (note that $k_1 \neq 0$). We find that $m_{\alpha'} \in \Gamma'$, otherwise, let $\alpha'' \in \mathbb{Z}_{\geq 0}^{\mathcal{A}'}$ such that $m_{\alpha''} = m_{\alpha'} + m_{\gamma} \in \Gamma \setminus \Gamma'$. Then the binomial corresponds to the new relation

$$(\alpha_0 + (k_1 - 1)\gamma + \alpha'') - (\beta_0 + \beta') \in L_{\mathcal{B}}$$

is not in $I_{\mathcal{A}} + I_{\mathcal{A}'}$ by (4.15), which leads to a contradiction of the minimality of k_1 . Note that $m_{\alpha_0 + k_1\gamma} \in S_{\sigma_0}$ since $m_{\alpha'} \in \bar{\Gamma}$ implies

$$\langle m_{\alpha_0 + k_1\gamma}, u_3 \rangle = \langle m_{\alpha_0 + k_1\gamma} + m_{\alpha'}, u_3 \rangle = \langle m_{\beta_0 + \beta'}, u_3 \rangle \geq 0.$$

Then there exists $\tilde{\alpha}_0 \in \mathbb{Z}_{\geq 0}^{\mathcal{A}_0}$ such that $m_{\tilde{\alpha}_0} = m_{\alpha_0 + k_1\gamma}$. Then we get the contradiction from

$$x^{\alpha_0 + k_1\gamma} - x^{\tilde{\alpha}_0} \in I_{\mathcal{A}}, \quad x^{\tilde{\alpha}_0 + \alpha'} - x^{\beta_0 + \beta'} \in I_{\mathcal{A}'} \implies x^{\alpha_0 + k_1\gamma + \alpha'} - x^{\beta_0 + \beta'} \in I_{\mathcal{A}} + I_{\mathcal{A}'}$$

and thus (iv') is true. Now we have finished the **proof of our claim**.

Finally, it suffices to show that (4.12) is equivalent to the condition (v). Given $m = (-p_1, p_2, p_3) \in \Gamma$, that is, $p_1, p_3 \geq 0$ and $-b_2p_2 \geq -b_1p_1 + b_3p_3$, we want to find a lattice point $m' = (-q_1, q_2, 0) \in \Gamma'$ such that $m - m' = (q_1 - p_1, p_2 - q_2, p_3) \in S_{\sigma_0}$, that is, there exist $q_1, q_2 \geq 0$ such that

$$-b_2q_2 \geq -b_1q_1, \quad q_1 \geq p_1, \quad q_2 \leq p_2, \quad k := b_1p_1 - b_2p_2 - b_3p_3 \geq -b_2q_2 + b_1q_1. \quad (4.16)$$

Note that if (q_1, q_2) satisfies (4.16), then $(q_1, \lceil \frac{b_1 q_1}{b_2} \rceil)$ satisfies (4.16). Thus, (4.16) is equivalent to the existence of an integer $q_1 \in [p_1, b_2 p_2 / b_1]$ such that

$$k \geq -b_2 \left\lceil \frac{b_1 q_1}{b_2} \right\rceil + b_1 q_1 = g \cdot \{-b'_1 q_1\}_{b'_2}.$$

Consider the following statement $P(p_1, p_2)$:

$$\text{there exists } p_1 \leq q_1 \leq \frac{b'_2 p_2}{b'_1} \text{ such that } \{g(b'_2 p_2 - b'_1 p_1)\}_{b'_3} \geq g \cdot \{-b'_1 q_1\}_{b'_2}.$$

Hence, (4.12) is also equivalent to $P(p_1, p_2)$ holds for all $p_1, p_2 > 0$ with $b'_2 p_2 \geq b'_1 p_1$. Note that if $b'_2 p_2 / b'_1 - p_1 \geq b'_2$, then there exists an integer $q_1 \in [p_1, b'_2 p_2 / b'_1]$ such that $b'_2 \lfloor q_1 \rfloor$, ensuring that the above inequality holds. Therefore, we only need to check it for $\lambda := b'_2 p_2 - b'_1 p_1 \leq b'_1 b'_2$. We take $n_1, n_2 \in \mathbb{N}$ such that $b'_2 n_2 - b'_1 n_1 = 1$. Since $P(p_1, p_2)$ holds if and only if $P(p_1 + b'_2, p_2 + b'_1)$ holds, we only need to check that $(p_1, p_2) = (\lambda n_1, \lambda n_2)$ for $0 \leq \lambda \leq b'_1 b'_2$. Let $q_1 = \lambda n_1 + y$. Clearly, $P(\lambda n_1, \lambda n_2)$ can be reformulated as the existence of

$$0 \leq y \leq \frac{b'_2 \cdot \lambda n_2}{b'_1} - \lambda n_1 = \frac{\lambda}{b'_1}$$

such that

$$\{g\lambda\}_b \geq g \cdot \{-b'_1(\lambda n_0 + y)\}_{b'_2} = g \cdot \{\lambda - b'_1 y\}_{b'_2}.$$

This establishes the equivalence (iv) $\Leftrightarrow$ (v). $\square$

Remark 4.6 Notice that the condition (v) is symmetric with respect to b_1 and b_2 . Indeed, given $0 \leq \lambda \leq b'_1 b'_2$, let $0 \leq y_1 \leq \lambda / b'_1$ satisfy the inequality (4.11), that is, there exists $0 \leq y_2 \leq \lambda / b'_2$ such that

$$\lambda - y_1 b'_1 - y_2 b'_2 = \{\lambda - y_1 b'_1\}_{b'_2}.$$

Then the inequality

$$\{\lambda - y_2 b'_2\}_{b'_1} = \{\{\lambda - y_1 b'_1\}_{b'_2}\}_{b'_1} \leq \{\lambda - y_1 b'_1\}_{b'_2} \leq \{g\lambda\}_{b_3}$$

as required. Hence we conclude that U_{31} is reduced if and only if U_{32} is reduced.

In fact, we can achieve a stronger result.

Theorem 4.7 X is reduced if and only if U_{31} is reduced.

This result is based on the following lemma.

Lemma 4.8 Let $\pi : X = X_\Sigma \times_{X_{\Sigma_0}} X_{\Sigma'} \rightarrow X_\Sigma$ be the first projection and $\mathcal{I}$ be the nilradical ideal of $\mathcal{O}_X$. Then the following vanishing results hold for the 3-dimensional case:

$$\pi_* \mathcal{I} = R^i \pi_* \mathcal{I} = R^i \pi_* \mathcal{O}_X = 0$$

for all $i > 0$, and $\pi_* \mathcal{O}_X = \mathcal{O}_{X_\Sigma}$.

Also, the similar results hold for the second projection $\pi' : X \rightarrow X_{\Sigma'}$.

Proof Since π is a proper morphism with fiber dimension ≤ 1 , by the formal function theorem, $R^i \pi_* \mathcal{I} = R^i \pi_* \mathcal{O}_X = 0$ for all $i > 1$. For $i = 1$, it suffices to show that

$$H^1(\pi^{-1}(U_{\sigma_j}), \mathcal{I}) = H^1(\pi^{-1}(U_{\sigma_j}), \mathcal{O}_X) = 0, \text{ for } j = 3, 4.$$

Since $\pi^{-1}(U_{\sigma_j})$ is covered by $\{U_{jk} := U_{\sigma_j} \times_{U_{\sigma_0}} U_{\sigma_k}\}_{k=1,2}$, by Čech cohomology, it suffices to show that

$$H^0(U_{j1}, \mathcal{J}) \oplus H^0(U_{j2}, \mathcal{J}) \longrightarrow H^0(U_{j1} \cap U_{j2}, \mathcal{J}) = H^0(U_{\sigma_j} \times_{U_{\sigma_0}} U_{\sigma_1 \cap \sigma_2}, \mathcal{J}) \quad (4.17)$$

$$H^0(U_{j1}, \mathcal{O}_X) \oplus H^0(U_{j2}, \mathcal{O}_X) \longrightarrow H^0(U_{j1} \cap U_{j2}, \mathcal{O}_X) = H^0(U_{\sigma_j} \times_{U_{\sigma_0}} U_{\sigma_1 \cap \sigma_2}, \mathcal{O}_X) \quad (4.18)$$

are surjective.

By the relation $b_1 u_1 + b_2 u_2 + b_3 u_3 + b_4 u_4 = 0$, we have $(\sigma_1 \cap \sigma_2)^\vee = \sigma_1^\vee \cup \sigma_2^\vee$, and thus

$$0 \rightarrow \mathbb{C}[\sigma_1^\vee \cap \sigma_2^\vee \cap M] \longrightarrow \mathbb{C}[\sigma_1^\vee \cap M] \oplus \mathbb{C}[\sigma_2^\vee \cap M] \longrightarrow \mathbb{C}[(\sigma_1 \cap \sigma_2)^\vee \cap M] \rightarrow 0 \\ (\chi^{m_1}, \chi^{m_2}) \longmapsto \chi^{m_1} - \chi^{m_2} \quad (4.19)$$

Since (4.18) is equal to the last morphism in (4.19) tensored by $\mathbb{C}[\sigma_j^\vee \cap M]$ over $\mathbb{C}[\sigma_0^\vee \cap M]$, we conclude that (4.18) is surjective.

We modify the notation in Theorem 3.1 as follows:

$$S_{\sigma_0} = \mathbb{Z}_{\geq 0} \mathcal{A}_0, \quad S_{\sigma_j} = \mathbb{Z}_{\geq 0} \mathcal{A}, \quad S_{\sigma_k} = \mathbb{Z}_{\geq 0} \mathcal{A}'_k \quad \text{for } k = 1, 2.$$

Since $S_{\sigma_1 \cap \sigma_2} = S_{\sigma_1} + S_{\sigma_2} = \mathbb{Z}_{\geq 0}(\mathcal{A}'_1 \cup \mathcal{A}'_2)$ and $\sigma_0 = \sigma_j + (\sigma_1 \cap \sigma_2)$, according to Remark 4.4 we get the equality

$$\sqrt{I_{\mathcal{A}} + I_{\mathcal{A}'_1 \cup \mathcal{A}'_2}} = I_{\mathcal{A} \cup (\mathcal{A}'_1 \cup \mathcal{A}'_2)}$$

in $\mathbb{C}[\mathbb{Z}_{\geq 0}^{\mathcal{A} \cup \mathcal{A}'_1 \cup \mathcal{A}'_2}]$, and thus

$$H^0(U_{\sigma_j} \times_{U_{\sigma_0}} U_{\sigma_1 \cap \sigma_2}, \mathcal{J}) = I_{\mathcal{A} \cup \mathcal{A}'_1 \cup \mathcal{A}'_2} / I_{\mathcal{A}} + I_{\mathcal{A}'_1 + \mathcal{A}'_2}$$

is generated by $\overline{x^{\alpha+\alpha'} - x^{\beta+\beta'}}$, where $\alpha, \beta \in \mathbb{Z}_{\geq 0}^{\mathcal{A} \setminus \mathcal{A}'_0}$, $\alpha', \beta' \in \mathbb{Z}_{\geq 0}^{\mathcal{A}'_1 \cup \mathcal{A}'_2}$ with $(\alpha + \alpha') - (\beta + \beta') \in L_{\mathcal{A} \cup \mathcal{A}'_1 \cup \mathcal{A}'_2}$.

If $m_{\alpha'}, m_{\beta'} \in S_{\sigma_k}$ for some $k \in \{1, 2\}$, say $\alpha'', \beta'' \in \mathbb{Z}_{\geq 0}^{\mathcal{A}'_k}$ such that $m_{\alpha'} = m_{\alpha''}$ and $m_{\beta'} = m_{\beta''}$, then

$$x^{\alpha+\alpha'} - x^{\beta+\beta'} \equiv x^{\alpha+\alpha''} - x^{\beta+\beta''} \pmod{I_{\mathcal{A}'_1 + \mathcal{A}'_2}},$$

and $(-1)^{k-1} \overline{x^{\alpha+\alpha''} - x^{\beta+\beta''}} \in I_{\mathcal{A} \cup \mathcal{A}'_k} / (I_{\mathcal{A}} + I_{\mathcal{A}'_k})$ maps to $\overline{x^{\alpha+\alpha'} - x^{\beta+\beta'}} \in I_{\mathcal{A} \cup \mathcal{A}'_1 \cup \mathcal{A}'_2} / (I_{\mathcal{A}} + I_{\mathcal{A}'_1 + \mathcal{A}'_2})$.

If not, we may assume that $m_{\alpha'} \in S_{\sigma_1}$ and $m_{\beta'} \in S_{\sigma_2}$. Since

$$m_{\alpha} + m_{\alpha'} = m_{\beta} + m_{\beta'} \in (S_{\sigma_j} + S_{\sigma_1}) \cap (S_{\sigma_j} + S_{\sigma_2}) \subseteq S_{\sigma_j \cap \sigma_1} \cap S_{\sigma_j \cap \sigma_2} \subseteq S_{\sigma_j},$$

there exist $\gamma_0 \in \mathbb{Z}_{\geq 0}^{\mathcal{A}_0}$ and $\gamma \in \mathbb{Z}_{\geq 0}^{\mathcal{A} \setminus \mathcal{A}'_0}$ such that

$$m_{\gamma_0} + m_{\gamma} = m_{\alpha} + m_{\alpha'} = m_{\beta} + m_{\beta'}.$$

Then

$$\overline{(x^{\alpha+\alpha'} - x^{\gamma_0+\gamma}, x^{\beta+\beta'} - x^{\gamma_0+\gamma})} \mapsto \overline{x^{\alpha+\alpha'} - x^{\beta+\beta'}},$$

under the morphism $H^0(U_{j1}, \mathcal{O}_X) \oplus H^0(U_{j2}, \mathcal{O}_X) \rightarrow H^0(U_{j1} \cap U_{j2}, \mathcal{O}_X)$. Hence, we conclude that (4.17) is surjective.

Note that

$$0 \rightarrow \pi_* \mathcal{I} \rightarrow \pi_* \mathcal{O}_X \rightarrow \pi_* \mathcal{O}_{X_{\text{red}}} \rightarrow R^1 \pi_* \mathcal{I} = 0. \quad (4.20)$$

By Corollary 3.3 and Theorem 4.3, $X_{\text{red}} = X_{\Sigma}$ is integral. By the proof of Zariski main theorem in [4], $\pi_* \mathcal{O}_{X_{\text{red}}} = \mathcal{O}_{X_{\Sigma}}$. By functoriality, the morphism between structure sheaves $\mathcal{O}_{X_{\Sigma}} \rightarrow \pi_* \mathcal{O}_X$ gives a lifting of (4.20), and thus the short exact sequence (4.20) splits.

Now the remaining part is to prove that $\pi_* \mathcal{I} = 0$. By definition and $R^1 \pi_* \mathcal{O}_X = 0$, we have

$$0 \rightarrow \mathcal{O}_X(\pi^{-1} U_{\sigma_j}) \rightarrow \mathcal{O}_X(U_{j1}) \oplus \mathcal{O}_X(U_{j2}) \rightarrow \mathcal{O}_X(U_{\sigma_j} \times_{U_{\sigma_0}} U_{\sigma_1 \cap \sigma_2}) \rightarrow 0,$$

or rewrite it as

$$\begin{aligned} 0 \rightarrow \mathbb{C}[\mathcal{S}_{\sigma_j}] \oplus \Gamma(\pi_* \mathcal{I}, U_{\sigma_j}) &\rightarrow \mathbb{C}[\mathcal{S}_{\sigma_j}] \otimes_{\mathbb{C}[\mathcal{S}_{\sigma_0}]} (\mathbb{C}[\mathcal{S}_{\sigma_1}] \oplus \mathbb{C}[\mathcal{S}_{\sigma_2}]) \\ &\xrightarrow{\alpha} \mathbb{C}[\mathcal{S}_{\sigma_j}] \otimes_{\mathbb{C}[\mathcal{S}_{\sigma_0}]} \mathbb{C}[\mathcal{S}_{\sigma_1 \cap \sigma_2}] \rightarrow 0, \end{aligned}$$

since $\pi_* \mathcal{O}_X = \pi_* \mathcal{I} \oplus \mathcal{O}_{X_{\Sigma}}$. From the long exact sequence induced by (4.19) $\otimes_{\mathbb{C}[\mathcal{S}_{\sigma_0}]} \mathbb{C}[\mathcal{S}_{\sigma_j}]$, the kernel of α is

$$\mathbb{C}[\mathcal{S}_{\sigma_j}] / \text{Im} \left(\text{Tor}_1^{\mathbb{C}[\mathcal{S}_{\sigma_0}]} (\mathbb{C}[\mathcal{S}_{\sigma_1 \cap \sigma_2}], \mathbb{C}[\mathcal{S}_{\sigma_j}]) \rightarrow \mathbb{C}[\mathcal{S}_{\sigma_j}] \right).$$

So we can conclude that $\Gamma(\pi_* \mathcal{I}, U_{\sigma_j}) = 0$. Hence $\pi_* \mathcal{I}|_{U_{\sigma_j}} = 0$ for $j = 1, 2$, that is, $\pi_* \mathcal{I} = 0$. $\square$

Proof of Theorem 4.7 Suppose that U_{31} is reduced. Recall that $J_- = \{1, 2\}$ and $J_+ = \{3, 4\}$. Let $Z = V(\sigma_{J_-})$, $Z' = V(\sigma_{J_+})$ and $S = V(\sigma_{J_- \cup J_+})$. By Remark 4.6, U_{32} is also reduced, and thus

$$\text{Supp } \mathcal{I} \subseteq (Z \times_S Z')_{\text{red}} \setminus (U_{31} \cup U_{32}) = (\pi')^{-1}(p),$$

where p is the unique point in $V(\sigma_1 \cap \sigma_2) \setminus U_{\sigma_1} = V(\sigma_2)$. Note that $(Z \times_S Z')_{\text{red}} = \mathbb{P}^1 \times \mathbb{P}^1$, and $\pi|_{\mathbb{P}^1 \times \mathbb{P}^1}$ and $\pi'|_{\mathbb{P}^1 \times \mathbb{P}^1}$ are projection onto each component. We take a section $s: Z = \mathbb{P}^1 \xrightarrow{\sim} (\pi')^{-1}(p)_{\text{red}} = \mathbb{P}^1$ such that $\pi \circ s = \text{id}_Z$. Let $\mathcal{F} = (s^{-1})_* \iota'^{-1} \mathcal{I}$ be the sheaf of abelian groups on Z , where $\iota': (\pi')^{-1}(p)_{\text{red}} \hookrightarrow X$. Since $\mathcal{I}$ has the support on $(\pi')^{-1}(p)_{\text{red}}$ and by Lemma 4.8, we have

$$\iota_* \mathcal{F} = \pi_* \iota'_* s_* (s^{-1})_* \iota'^{-1} \mathcal{I} = \pi_* \mathcal{I} = 0,$$

where $\iota: Z \hookrightarrow X_{\Sigma}$. This implies $\mathcal{F} = 0$, and thus $\mathcal{I} = 0$, i.e., X is reduced. $\square$

5 Applications

5.1 About terminal and canonical singularities

Let $\phi: X_{\Sigma} \rightarrow X_{\Sigma_0}$ be the flipping contraction of an extremal ray $\mathcal{R}$ and $\phi': X_{\Sigma'} \rightarrow X_{\Sigma_0}$ be its corresponding flip. We can apply the results in Sect. 4 to study $X = X_{\Sigma} \times_{X_{\Sigma_0}} X_{\Sigma'}$ when X_{Σ} is a 3-dimensional simplicial toric variety with at worst terminal or canonical singularities.

Theorem 5.1 *Let X_{Σ} be a 3-dimensional simplicial toric variety with at worst terminal singularities. If $K_{X_{\Sigma}} \cdot \mathcal{R} \leq 0$, then the fiber product $X = X_{\Sigma} \times_{X_{\Sigma_0}} X_{\Sigma'}$ is the toric variety X_{Σ} defined by the fan (2.6).*

Proof We have known that X is irreducible by Theorem 4.3. Without loss of generality, we may assume that X_{Σ_0} is affine. By the classification result for 3-dimensional terminal singularities [2, Corollary 2.1, Theorem 3.1] (see also [7, 14-2-5]), the wall relation for $K_{X_\Sigma} \cdot \mathcal{R} < 0$ is given by

$$-au_1 - (r - a)u_2 + ru_3 + u_4 = 0 \quad \text{or} \quad -au_1 - u_2 + ru_3 + u_4 = 0,$$

and for $K_{X_\Sigma} \cdot \mathcal{R} = 0$ is given by

$$-u_1 - u_2 + u_3 + u_4 = 0,$$

where $\{u_1, u_2, u_3\}$ is a $\mathbb{Z}$ -basis of $N = \mathbb{Z}^3$, $0 < a < r$ and $\gcd(a, r) = 1$. In particular, the toric variety X_Σ satisfies Corollary 3.3 (1) and thus the graph closure $\overline{\Gamma}_f$ is normal, where $f = (\phi')^{-1} \circ \phi$.

To see the reduced property of X , by Theorem 4.7, it suffices to show that the numerical criterion (v) in Lemma 4.5 holds. For the first case, we have $g = 1$. Given $\lambda \geq 0$, we can take $y = \lfloor \frac{\lambda}{r} \rfloor \leq \frac{\lambda}{a}$. Then

$$\lambda - ay = ry + \{\lambda\}_r - ay \implies \{\lambda - ay\}_{r-a} = \{\{\lambda\}_r\}_{r-a} \leq \{g\lambda\}_r.$$

For the second and third cases, given any $\lambda \geq 0$ we can take $y = 0$ since $b'_2 = 1$. Therefore X is a normal toric variety by Proposition 4.1. $\square$

We now assume for simplicity that $X_{\Sigma_0} = U_{\sigma_0}$. Recall that the extremal ray $\mathcal{R}$ is defined by the wall relation

$$\sum_{i \in J_-} b_i u_i + \sum_{j \in J_+} b_j u_j = 0. \quad (5.1)$$

After suitable scaling, we may assume that $b_i \in \mathbb{Z}$ for $i \in J_- \cup J_+$ and $\gcd(b_1, \dots, b_{n+1}) = 1$.

The following proposition illustrates the fiber product X may be not reduced when X_Σ has canonical singularities (see Remark 5.3).

Proposition 5.2 Suppose that $\{u_1, u_2, u_3\}$ is a $\mathbb{Z}$ -basis of $N \simeq \mathbb{Z}^3$, and $J_- = \{1, 2\}$, $J_+ = \{3, 4\}$ and $b_4 = 1$. Assume further that

$$b_1 + b_2 + b_3 + 1 = 0. \quad (5.2)$$

Then the fiber product X is the toric variety $X_{\tilde{\Sigma}}$ if and only if there exist two non-negative integers y_1 and y_2 such that

$$b_3 = b'_1 y_1 + b'_2 y_2 \quad (5.3)$$

where $g = \gcd(b_1, b_2) > 0$ and $b_i = -gb'_i$ for $i = 1, 2$.

Proof The irreducibility of X and the normality of $\overline{\Gamma}_f$ follow as in the proof of Theorem 5.1. By Theorem 4.7 and Proposition 4.1, the proposition follows from the claim that (5.3) is equivalent to (v) in Lemma 4.5.

For (v) $\Rightarrow$ (5.3), according to the proof in Lemma 4.5, the statement (v) in Lemma 4.5 holds for all $\lambda \geq 0$. In particular, for $\lambda = b_3$, there exists $0 \leq y_1 \leq b/b'_1$ such that

$$\{b_3 - b'_1 y_1\}_{b'_2} \leq \{gb_3\}_{b_3} = 0.$$

That is, there exists $y_2 \in \mathbb{Z}_{\geq 0}$ with $b_3 - b'_1 y_1 = b'_2 y_2$ as required.

For (5.3) $\Rightarrow$ (v), let $\lambda = pb_3 + q(b'_1 + b'_2) + r$ where

$$0 \leq q \leq g - 1 \quad \text{and} \quad 0 \leq r \leq b'_1 + b'_2 - 1 - \delta_{q, g-1}.$$

We take $y = py_1 + q \leq pb_3/b'_1 + q \leq \lambda/b'_1$, and notice that

$$\{g\lambda\}_{b_3} = \{q(b_1 + b_2) + gr\}_{b_3} = \{q + gr\}_{b_3} = q + gr,$$

since $0 \leq q + gr \leq g - 1 + g(b'_1 + b'_2 - 1 - \delta_{q,g-1}) < b_3$. On the other hand,

$$\{\lambda - b'_1 y\}_{b'_2} = \{p(b_3 - b'_1 y_1) + qb'_1 + r - qb'_1\}_{b'_2} = \{r\}_{b'_2}.$$

Hence

$$\{g\lambda\}_{b_3} = q + gr \geq g \cdot \{r\}_{b'_2} = g \cdot \{\lambda - b'_1 y\}_{b'_2}.$$

□

Remark 5.3 Note that the singularities of X_Σ in Proposition 5.2 are at worst canonical singularities. Indeed, since σ_4 is smooth cone, we only need to check the condition (b) in [1, Proposition 11.4.12] for σ_3 , where $\sigma_i = \sigma_{\{1,2,3,4\} \setminus \{i\}}$. Set the polytope $\Pi_{\sigma_3} = \text{Conv}(0, u_1, u_2, u_4)$. For $m \in \Pi_{\sigma_3} \cap M$, there are $a, a_i \in \mathbb{Z}_{\geq 0}$ and $a_1 + a_2 + a_3 \leq a$ such that $am = a_1 u_1 + a_2 u_2 + a_3 u_4$. Then

$$a \mid (a_1 - b_1 a_3), (a_2 - b_2 a_3), -a_3 b_3,$$

since $\{u_1, u_2, u_3\}$ is a $\mathbb{Z}$ -basis and the wall relation (5.1), and thus

$$a_1 + a_2 + a_3 = a_1 + a_2 - a_3(b_1 + b_2 + b_3) \equiv 0 \pmod{a}.$$

This implies that $m \in \text{Conv}(u_1, u_2, u_4) \cup \{0\}$, since $0 \leq a_1 + a_2 + a_3 \leq a$.

Also, $X_\Sigma \rightarrow U_{\sigma_0}$ is a flopping contraction since $K_{X_\Sigma} \cdot \mathcal{R} = 0$ by (5.2).

Example 5.4 By the elementary number theory, if $b_3 \geq (b'_1 - 1)(b'_2 - 1)$, then the condition (5.3) will hold. In this case, the fiber product X is the toric variety by Proposition 5.2.

On the other hand, taking $(b_1, b_2) = (-3, -3k - 5)$ for $k \in \mathbb{Z}_{\geq 0}$, we find that the fiber product X is not a toric variety in this case. In fact, it is easily seen that if (b_1, b_2) does not satisfy (5.3) then $(b_1, b_1 + b_2)$ also does not satisfy (5.3).

5.2 Higher dimensional case

Since the criterion for being reduced of X only holds for 3-dimensional case, it seems difficult to say something more about higher dimensional case. However, if we assume the smoothness of X_Σ , then by more detailed argument, a useful criterion can also be achieved (Fig 2).

Theorem 5.5 Assume that X_Σ is smooth of dimension n and X_{Σ_0} is affine with the wall relation (5.1). Then the fiber product $X = X_\Sigma \times_{X_{\Sigma_0}} X_{\Sigma'}$ is the toric variety $X_{\tilde{\Sigma}}$ if and only if

$$b_i \mid b_j \text{ or } b_j \mid b_i \quad (5.4)$$

for any $i, j \in J_-$.

Proof The irreducibility of X and the normality of $\overline{\Gamma}_f$ are true by Theorem 4.3 and Corollary 3.3 (2). Then by Proposition 4.1, the result follows from the claim that (5.4) is equivalent to the generalized version of (4.12) in Lemma 4.5 for any affine piece of X .

Recall that $\gcd(b_1, \dots, b_{n+1}) = 1$. Given $j \in J_+$, we have that the set $\{u_k \mid k \neq j\}$ forms a $\mathbb{Z}$ -basis of N since X_Σ is smooth, so the wall relation (5.1) implies that $b_j \mid b_k$ for each $1 \leq k \leq n + 1$, and thus $b_j = 1$ for each $j \in J_+$.

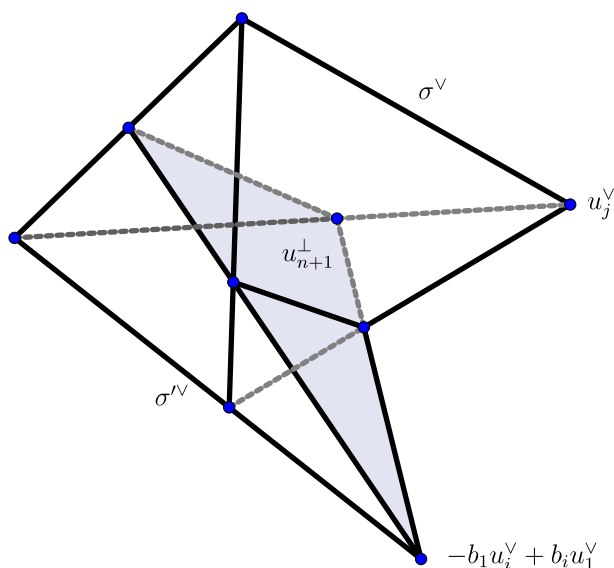


Fig. 2 $\sigma^\vee$ and $(\sigma')^\vee$ which are cut by a hyperplane in M

For simplicity, we assume that $1 \in J_-$ and $n+1 \in J_+$. Let $J_+^* = J_+ \setminus \{n+1\}$ and $J_-^* = J_- \setminus \{1\}$. We will prove that $b_1 \mid b_i$ or $b_i \mid b_1$ for each $i \in J_-^*$ if and only if $U_\sigma \times_{U_{\sigma_0}} U_{\sigma'}$ is reduced, where $\sigma = \sigma_{n+1}$ and $\sigma' = \sigma_1$.

Assume that $U_\sigma \times_{U_{\sigma_0}} U_{\sigma'}$ is reduced. Observe that $u_j^\vee \in \sigma^\vee \setminus \sigma'^\vee$ spans a ray of $\sigma^\vee$ for each $j \in J_+^*$ as presented in Fig 2, where $\{u_1^\vee, \dots, u_n^\vee\}$ is the dual basis of $\{u_1, \dots, u_n\}$. Notice that the generating set $\mathcal{A}$ can be selected to be $\mathcal{A}_0 \cup \{u_j^\vee \mid j \in J_+^*\}$, since σ is smooth. Set

$$\Gamma = M \cap \overline{\sigma'^\vee \setminus \sigma^\vee} \quad \text{and} \quad \Gamma' = \Gamma \cap \text{Cone}(u_{n+1})^\perp.$$

Then $\langle z, u_{n+1} \rangle > 0$ for each $z \in \Gamma \setminus \Gamma'$. Since $\langle u_j^\vee, u_{n+1} \rangle = -1$, we still have $\langle z + u_j^\vee, u_{n+1} \rangle \geq 0$ and thus $z + u_j^\vee \in \Gamma$ for all $j \in J_+^*$. By the same argument in Lemma 4.5, if $U_\sigma \times_{U_{\sigma_0}} U_{\sigma'}$ is reduced, then

$$\Gamma \subseteq S_{\sigma_0} + \Gamma'. \quad (5.5)$$

Let $\Gamma'' = \Gamma' \cap \bigcap_{j \in J_+^*} \text{Cone}(u_j)^\perp$. For any $z \in \Gamma' \setminus \Gamma''$, there is a $j_1 \in J_+^*$ such that $\langle z, u_{j_1} \rangle > 0$. If $|J_+| \neq 2$, then we can choose another index $j' \in J_+^* \setminus \{j_1\}$, which implies that

$$(z - u_{j_1}^\vee + u_{j'}^\vee) + u_{j_1}^\vee = z + u_{j'}^\vee$$

and $z - u_{j_1}^\vee + u_n^\vee \in \Gamma'$. Again, by the same argument, we have

$$\Gamma' \setminus \Gamma'' \subseteq (S_{\sigma_0} \setminus 0) + \Gamma'' \quad (5.6)$$

if $U_\sigma \times_{U_{\sigma_0}} U_{\sigma'}$ is reduced. Combining (5.5) and (5.6), we get

$$\Gamma \subseteq S_{\sigma_0} + \Gamma''. \quad (5.7)$$

Conversely, we claim that (5.7) (resp. (5.5)) implies that $U_\sigma \times_{U_{\sigma_0}} U_{\sigma'}$ is reduced when $|J_+| \neq 2$ (resp. $|J_+| = 2$). As in the proof of Theorem 3.1, we pick

$$(\alpha_0 + \alpha + \alpha') - (\beta_0 + \beta + \beta') \in L_{\mathcal{B}} \quad (5.8)$$

where $\alpha_0, \beta_0 \in \mathbb{Z}_{\geq 0}^{\mathcal{A}_0}$, $\alpha, \beta \in \mathbb{Z}_{\geq 0}^{\mathcal{A}' \setminus \mathcal{A}_0}$ and $\alpha', \beta' \in \mathbb{Z}_{\geq 0}^{\mathcal{A}' \setminus \mathcal{A}_0}$. If $|J_+| \neq 2$, by the decomposition (5.7), then we may assume that $\alpha', \beta' \in \Gamma''$. Since $\tilde{S}_\sigma$ is generated by $\{u_i^\vee \mid 1 \leq i \leq n\}$ as a semigroup, we may assume that $\alpha_0, \beta_0 \in \text{Cone}(u_i^\vee \mid i \in J_- \cup J_0)$. In this case, since

$$\alpha_0, \beta_0, \alpha', \beta' \in \bigcap_{j \in J_+^*} u_j^\perp \quad \text{and} \quad \alpha, \beta \in \bigcap_{i \in (J_- \cup J_0)} u_i^\perp,$$

we conclude that $\alpha = \beta$ and thus

$$x^{\alpha_0 + \alpha + \alpha'} - x^{\beta_0 + \beta + \beta'} = x^\alpha (x^{\alpha_0 + \alpha'} - x^{\beta_0 + \beta'}) \in I_{\mathcal{A}} + I_{\mathcal{A}'},$$

If $|J_+| = 2$, then $\alpha, \beta \in \mathbb{Z}_{\geq 0} \text{Cone}(u_\ell)^\perp$ where $J_+ = \{\ell, n+1\}$. Then the same argument in Lemma 4.5 will work.

We can translate the condition (5.7) (resp. (5.5)) to the following condition:

(♠) Given $z = \sum_{i=1}^n z_i u_i^\vee \in \Gamma$, that is, $z_1 \leq 0, z_i \geq 0$ for $i \neq 1$ and

$$\sum_{i=1}^n b_i z_i \leq 0,$$

there exists $z' = \sum_{i \in (J_- \cup J_0)} z'_i u_i^\vee \in \Gamma''$ (resp. $z' = \sum_{i=1}^n z'_i u_i^\vee \in \Gamma'$) such that $z - z' \in S_{\sigma_0}$, that is, there exists $z'_1 \leq 0, z'_i \geq 0$ for all $i \neq 1$ such that

$$\sum_i b_i z'_i = 0,$$

and $z_i \geq z'_i$ for all i .

We claim that if (♠) holds, then $b_1 \mid b_i$ or $b_i \mid b_1$ for each $i \in J_-^*$. Indeed, let $g = \gcd(b_1, b_i) > 0$ and $b_1 = -gb'_1, b_i = -gb'_i$. There exists integers $0 \leq n_1 < b'_i$ and n_i such that $b'_i n_i = b'_1 n_1 + 1$. Note that

$$n_i = \frac{b'_1 n_1 + 1}{b'_i} \leq \frac{b'_1(b'_i - 1) + 1}{b'_i} = b'_1 + \frac{1 - b'_1}{b'_i} \leq b'_1 \quad (5.9)$$

and the equality holds only when $b'_1 = 1$ and $n_1 = b'_i - 1$. Consider $z = -n_1 u_1^\vee + n_i u_i^\vee \in \bar{\Gamma}$. Then there exists $z'_i \geq 0 \geq z'_1$ such that $-n_1 \geq z'_1$ and $n_i \geq z'_i$ by (♠) and $b'_1 z'_1 + b'_i z'_i = 0$. If $b'_i \neq 1$, then $n_1 > 0$. Since $b'_i \mid z'_1$ and $z'_1 \leq -n_1 < 0$, we must have $z'_1 \leq -b'_i$, and thus

$$b'_1 \geq n_i \geq z'_i = \frac{-b_1 z'_1}{b'_i} \geq b'_1.$$

We conclude that $n_i = b'_1$, and thus $b'_1 = 1$, since the equality in (5.9) holds.

Conversely, we will show that (5.4) implies (♠). Given $z \in \Gamma$, we define $\tilde{z} = \sum_{i \in J_-} \tilde{z}_i u_i^\vee$ by setting

$$\tilde{z}_1 = \left\lfloor \frac{-1}{b_1} \sum_{i \in J_-^*} b_i z_i \right\rfloor \quad \text{and} \quad \tilde{z}_i = z_i \quad \text{for } i \in J_-^*.$$

Then $\sum_{i \in J_-} b_i \tilde{z}_i \leq 0$ by construction, so that $\tilde{z} \in \Gamma$. If there exists $z' = \sum_{i \in J_-} z'_i u_i^\vee \in \Gamma'' \subseteq \Gamma'$ such that $\tilde{z} - z' \in S_{\sigma_0}$, then

$$z - z' = (z - \tilde{z}) + (\tilde{z} - z') \in \mathbb{Z}_{\geq 0} u_1^\vee + S_{\sigma_0} \subseteq S_{\sigma_0}.$$

So it suffices to show that

(♠') Given $z = \sum_{j \in J_-} z_1 u_j^\vee$, where $z_i \geq 0$ for $i \in J_-^*$ and $z_1 = \lfloor \frac{-1}{b_1} \sum_{j \in J_-} b_j z_j \rfloor$, there exists $z' = \sum_{i \in J_-} z'_i u_i^\vee \in \Gamma''$ such that $z - z' \in S_{\sigma_0}$.

Observe that condition (♠') holds when $b_i = -1$ for some $i \in J_-$. Indeed, we can take

$$z'_i = \sum_{k \in J_- \setminus \{i\}} b_k z_k \leq z_i \quad \text{and} \quad z'_k = z_k \quad \text{for } k \neq J_- \setminus \{i\}.$$

Moreover, since we can divide the greatest common divisor of $\{b_i\}_{i \in J_-}$ in the condition (♠) and go back to the case when $b_i = -1$ for some $i \in J_-$, we have (5.4) implies (♠'), and thus implies (♠). Hence X is reduced when (5.4) holds. $\square$

Appendix A: Remarks on K -equivalent toric varieties

Let $X = X_\Sigma$ and $X' = X_{\Sigma'}$ be two simplicial toric varieties with at most terminal singularities such that X and X' are K -equivalent, denoted by $X =_K X'$. We know that this is equivalent to that $\text{shed } \Sigma = \text{shed } \Sigma'$, or in other words, the fans Σ and Σ' give rise to different triangulations of the same polyhedron.

Let $f: X \dashrightarrow X'$ be a toric flop given by

$$\begin{array}{ccc} X = X_\Sigma & & X' = X_{\Sigma'} \\ & \searrow \phi_{\mathcal{R}} & \swarrow \phi' \\ & \bar{X} = X_{\Sigma_0} & \end{array}$$

and τ be a wall with $V(\tau) \in \mathcal{R}$ whose wall relation is $\sum_{i=1}^{n+1} b_i u_i = 0$. We may assume that $J_- = \{1, \dots, \alpha\}$, $J_+ = \{\beta + 1, \dots, n + 1\}$ and $b_{n+1} = 1$. Note that all the primitive vectors should all lie in an affine hyperplane of $N_{\mathbb{Q}}$ for toric flops. The exceptional set Z of $\phi_{\mathcal{R}}$ corresponding to $\text{Cone}(u_1, \dots, u_\alpha)$ under $\phi_{\mathcal{R}}$ of dimension $n - \alpha$ is mapped to $S := \phi_{\mathcal{R}}(Z) \subset \bar{X}$ corresponding to $\text{Cone}(u_1, \dots, u_\alpha, u_{\beta+1}, \dots, u_{n+1})$ of dimension $\beta - \alpha$. The map $Z \rightarrow S$ is a bundle with fiber covered by a weighted projective space $\tilde{\mathbb{P}}^{n-\beta}$ through a finite morphism (cf. [7], 14-2-3). Similar statements hold for $\phi': X' \rightarrow \bar{X}$ with the exceptional set Z' of ϕ' fibered over S with fiber covered by $\tilde{\mathbb{P}}^{\alpha-1}$.

Proposition A.1 *Any smooth toric flop is ordinary.*

Proof The smoothness condition tells us that the primitive generators $u_1, \dots, u_n$ form a $\mathbb{Z}$ -basis of the lattice N and so do $u_1, \dots, u_{n-1}, u_{n+1}$. When we represent u_{n+1} as a $\mathbb{Z}$ -linear combination of $u_1, \dots, u_n$ and u_n as a $\mathbb{Z}$ -linear combination of $u_1, \dots, u_{n-1}, u_{n+1}$ simultaneously, we can get that $b_i = -1$ for $i = 1, \dots, \alpha$ and $b_i = 1$ for $i = \beta + 1, \dots, n$.

All $u_1, \dots, u_n, u_{n+1}$ should all lie in an affine hyperplane of $N_{\mathbb{Q}}$. This implies that

$$-\sum_{i=1}^{\alpha} b_i = \sum_{i=\beta+1}^n b_i + 1$$

and thus $\alpha = n + 1 - \beta$.

Translating these data to the Reid's diagram, we have that $Z \rightarrow S$ is a bundle with fiber a projective space $\mathbb{P}^{n-\beta}$ and $Z' \rightarrow S$ is a bundle with fiber a projective space $\mathbb{P}^{\alpha-1}$. Note that $n - \beta = \alpha - 1$. It illustrates that the whole diagram for this case is an ordinary $\mathbb{P}^{\alpha-1}$ -flop. Hence we complete the proof. $\square$

By extending Reid's argument, we may decompose K -equivalent birational maps into toric flops.

Theorem A.2 *Let $X = X_\Sigma$ and $X' = X_{\Sigma'}$ be two simplicial toric varieties with at most terminal singularities such that $X =_K X'$. Then the birational map $f: X \dashrightarrow X'$ can be factorized into toric flops.*

Proof We know that $X =_K X'$ is equivalent to that the fans Σ and Σ' give rise to different simplicial triangulations of the same polyhedron fan $\bar{\Sigma}$. Also, the condition of having terminal singularities shows that the possible lattice points in $\text{shed } \Sigma$ and $\text{shed } \Sigma'$ are 0 and primitive generators of Σ and Σ' . From the above observation, we conclude that Σ and Σ' have the same edges and thus X and X' are isomorphic in codimension one.

Let $\bar{X} = X_{\bar{\Sigma}}$ and ϕ, ϕ' be the toric morphisms with respect to the simplicial triangulations of $\bar{\Sigma}$:

$$\begin{array}{ccc} X & & X' \\ & \searrow \phi & \swarrow \phi' \\ & \bar{X} & \end{array}$$

Also, we see easily that both ϕ and ϕ' are crepant morphisms.

We take an ample divisor H' in X' and consider H as the proper transform of H' in X . We run the $(K_X + H)$ -minimal model program of the morphism ϕ . Since K_X is ϕ -trivial and H is not ϕ -nef, we see that there must exist $(K_X + H)$ -extremal rays $\mathcal{R} = \mathbb{R}^+[C]$ with $\phi(C) = \text{pt}$ and $H \cdot C < 0$. This implies that the corresponding extremal contraction $\phi_{\mathcal{R}}$ is small. For if it is divisorial or fiber type, then we may represent $\mathcal{R}$ by an irreducible curve C which does not lie in the f exceptional loci $Z \subseteq X$. But since H is ample outside Z , we may make H and C transversal hence $H \cdot C \geq 0$, a contradiction.

Now we perform the toric flip $f_{\mathcal{R}}: X \dashrightarrow X^+$ of $\phi_{\mathcal{R}}$. Since $K_X \cdot C = 0$, $f_{\mathcal{R}}$ is indeed a toric flop and hence $f^+: X^+ \rightarrow \bar{X}$ still inherits all the relevant properties of X . In particular we still have $X^+ =_K X'$. By the termination of toric flips we conclude the proof of the theorem. $\square$

The termination of toric flips was stated with sketched proof in [7]. For the reader's convenience, we provide the details here.

Theorem A.3 *Let $X = X_\Sigma$ be a complete simplicial toric variety and $D = \sum_{\rho \in \Sigma(1)} a_\rho D_\rho$ be a $\mathbb{Q}$ -divisor with $0 \leq a_\rho < 1$. Then we have the termination of $(K_X + D)$ -flips.*

Proof Recall that $\text{shed } \Sigma = \{u \in \Sigma \mid \phi_{K_X}(u) \leq 1\}$ where ϕ_{K_X} is the corresponding piecewise linear function of K_X . Now we bring up a more general notion of relative shed of Σ with respect to D which is defined by the set $\{u \in \Sigma \mid \phi_{K_X+D}(u) \leq 1\}$ and is denoted by $\text{shed}_D \Sigma$. Our strategy is to give a similar geometric criterion about $\text{shed}_D \Sigma$ for the condition of $(K_X + D) \cdot V(\tau) < 0$ with τ a wall in Σ .

Indeed, when we restrict ourselves on an affine cone $\sigma = \text{Cone}(u_1, \dots, u_n)$ in Σ where u_i is the primitive vector of the 1-face ρ_i , $\text{shed}_D \Sigma|_\sigma$ is equal to the convex hull

$$\text{Conv} \left(0, \frac{u_1}{1 - a_{\rho_1}}, \dots, \frac{u_n}{1 - a_{\rho_n}} \right).$$

By the same argument in [8], we get that if $(K_X + D) \cdot V(\tau) < 0$ then $\text{shed}_D \Sigma$ has a bridge along τ and if $(K_X + D) \cdot V(\tau) > 0$ then $\text{shed}_D \Sigma$ has a gutter along τ . Hence if $X_{\Sigma'}$ is the $(K_X + D)$ -flipped variety, then

$$\text{Vol}(\text{shed}_D \Sigma) > \text{Vol}(\text{shed}_D \Sigma').$$

However the values of both volumes are in the discrete set

$$\left(\prod_{i=1}^n (1 - a_{\rho_i}) \right)^{-1} (n!)^{-1} \mathbb{N}.$$

This implies that it is impossible to have an infinite sequence of $(K_X + D)$ -flips. $\square$

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